

Hidden Cost of Your Dinner Plate: Which Foods Are Destroying the Planet?

1.0 Introduction

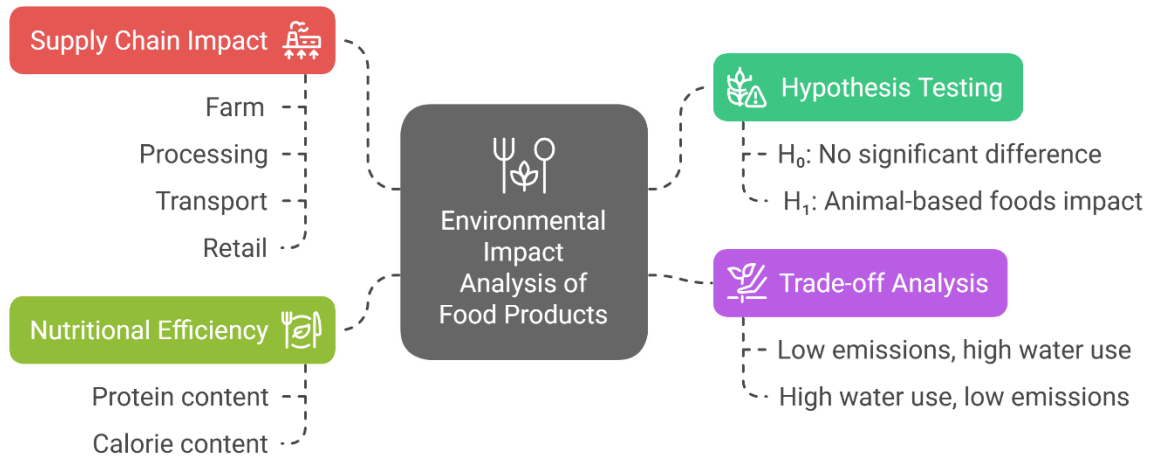


Figure 1. Conceptual framework illustrating the primary objectives and hypothesis of the food sustainability assessment. The diagram outlines how environmental impacts of animal-based and plant-based foods are evaluated across critical supply chain stages, trade-off considerations, and nutritional efficiency considering emissions, land use, water consumption, and pollution.

1.1 Background Information

The environmental impact of food production has emerged as a critical global concern, with agriculture contributing significantly to greenhouse gas (GHG) emissions, land degradation, freshwater depletion, and nutrient pollution. Research has shown that the expansion of agricultural activities has resulted in ecological damage such as deforestation, biodiversity loss, and water contamination, intensifying the urgency for sustainable food systems (Adedibu, 2023; Cimini, 2021). This project explores the environmental footprint of food production, comparing animal-based and plant-based foods in terms of their greenhouse gas emissions, land use, water consumption, and eutrophication potential. These factors are crucial to climate stability and ecological health.

Scientific studies have consistently demonstrated that plant-based diets generally impose lower environmental costs than animal-based diets, largely due to reduced emissions, smaller land requirements, and less intensive water use (Aleksandrowicz et al., 2016; Poore & Nemecek, 2018). This project builds on that foundation to deliver a robust, data-driven analysis that

evaluates food products and supply chain stages from production to retail. The analysis aims to identify the most environmentally burdensome food categories, detect trade-offs between different sustainability metrics, and guide actionable strategies toward more sustainable consumption and production (Clark et al., 2019; Springmann et al., 2016).

The primary objective is to quantify and compare the environmental impacts of food products, particularly distinguishing between animal-based and plant-based sources. This includes evaluating GHG emissions, land and water use, and pollution contributions per kilogram of food. Additional aims include analysing trade-offs where certain foods might have low emissions but high-water demand, or vice versa, and examining which stages of the supply chain such as farming, processing, or transport contribute most significantly to environmental degradation. The goal is to generate insights that inform sustainable decision-making among stakeholders.

These stakeholders include consumers interested in eco-conscious dietary choices, businesses and supply chain managers seeking efficiency and accountability in sourcing and logistics, and policymakers striving to create frameworks for sustainable agriculture and environmental protection. By making these insights accessible and actionable, the project contributes to the broader goal of reducing the food system's environmental burden while promoting health, sustainability, and resource efficiency across the value chain.

1.2 Key Factors in Environmental Impact Assessment of Food Products

This analysis focuses on the key environmental dimensions that influence the sustainability of food systems. Each metric represents a critical aspect of food production's impact on climate, land, water, and nutrient pollution. The study classifies foods into broader categories and subcategories for more structured comparisons, enabling actionable insights into the ecological burden of various food types.

Key categories assessed include:

- **Greenhouse Gas Emissions:** Evaluating emissions from all stages of the food supply chain.
- **Land Use:** Quantifying land requirements for different food products.
- **Water Consumption:** Measuring both total water withdrawals and stress-weighted impacts.

- **Eutrophication Potential:** Assessing nutrient runoff effects on freshwater ecosystems.
- **Supply Chain Breakdown:** Identifying stages such as farming, animal feed, transport, and packaging with the highest emissions.

To effectively assess these environmental dimensions, the following structured categories and features are used:

A. Food Product Classification

Food products are classified into animal-based and plant-based categories, with detailed subcategories to facilitate nuanced analysis.

Category	Subcategory	Food Products
Animal-Based Foods	Red Meat	Beef (beef herd), Beef (dairy herd), Lamb & Mutton, Pig Meat
	Poultry	Poultry Meat, Eggs
	Dairy	Milk, Cheese
	Seafood	Fish (farmed), Shrimps (farmed)
Plant-Based Foods	Cereal Grains & Staples	Wheat & Rye (Bread), Maize (Meal), Barley (Beer), Oatmeal, Rice, Potatoes, Cassava
	Legumes & Pulses	Other Pulses, Peas
	Nuts & Seeds	Nuts, Groundnuts
	Soy Products	Soymilk, Tofu
	Vegetable Oils	Soybean Oil, Palm Oil, Sunflower Oil, Rapeseed Oil, Olive Oil
	Vegetables	Onions & Leeks, Root Vegetables, Brassicas
	Fruits	Tomatoes, Citrus Fruit, Bananas, Apples, Berries & Grapes
	Beverages & Alcohol	Wine

Category	Subcategory	Food Products
Plant-Based Foods	Stimulants & Confectionery	Coffee, Dark Chocolate
	Sugars	Cane Sugar, Beet Sugar

- **Total Products:** 43
- **Animal-Based Foods:** 10
- **Plant-Based Foods:** 33

B. Environmental Feature Engineering

Feature names from the original dataset are standardized to improve readability, consistency, and analysis.

Category	Feature	Description	Units
GHG Emissions Metrics	Land_use_Change	GHG emissions from land use change	kg CO ₂ -eq/kg food
	Animal_Feed	Emissions from feed production	kg CO ₂ -eq/kg food
	Farm	Direct on-farm emissions	kg CO ₂ -eq/kg food
	Processing	Emissions from food processing	kg CO ₂ -eq/kg food
	Transport	Emissions from transporting food	kg CO ₂ -eq/kg food
	Packaging	Emissions from food packaging	kg CO ₂ -eq/kg food
	Retail	Emissions from refrigeration and storage	kg CO ₂ -eq/kg food

Category	Feature	Description	Units
GHG Emissions Metrics	Total_emissions	Combined GHG emissions from all stages	kg CO ₂ -eq/kg food
	GHG_1000kcal	GHG emissions per 1000 kcal	kg CO ₂ -eq/1000 kcal
	GHG_100gP	GHG emissions per 100g protein	kg CO ₂ -eq/100g protein
Eutrophication Metrics	Eutro_kg	Eutrophying emissions per kg of food	g PO ₄ -eq/kg
	Eutro_1000kcal	Eutrophying emissions per 1000 kcal	g PO ₄ -eq/1000 kcal
	Eutro_100gP	Eutrophying emissions per 100 g of protein	g PO ₄ -eq/100g protein
Water Use Metrics	Water_kg	Freshwater withdrawals per kg of food	L/kg
	Water_1000kcal	Freshwater withdrawals per 1000 kcal	L/1000 kcal
	Water_100gP	Freshwater withdrawals per 100 g protein	L/100g protein
Land Use Metrics	Land_kg	Land use per kg of food	m ² /kg
	Land_1000kcal	Land use per 1000 kcal	m ² /1000 kcal
	Land_100gP	Land use per 100 g of protein	m ² /100g protein
Scarcity-Weighted Water Use Metrics	SW_Water_kg	Scarcity-weighted water use per kg	L/kg
	SW_Water_1000kcal	Scarcity-weighted water use per 1000 kcal	L/1000 kcal
	SW_Water_100gP	Scarcity-weighted water use per 100 g of protein	L/100g protein

C. Summary of Core Metric Categories

Category	Features
GHG Emissions	Land_use_Change, Animal_Feed, Farm, Processing, Transport, Packaging, Retail, Total_emissions, GHG_1000kcal, GHG_100gP
Eutrophication	Eutro_kg, Eutro_1000kcal, Eutro_100gP
Water Use	Water_kg, Water_1000kcal, Water_100gP
Land Use	Land_kg, Land_1000kcal, Land_100gP
Water Scarcity Impact	SW_Water_kg, SW_Water_1000kcal, SW_Water_100gP

D. Metric Use Cases and Applicability

Each metric offers distinct insight depending on the analytical goal or context.

Metric Format	Use Case	Example Question
Per kg of food	Assess absolute impact per unit weight	“Which foods generate the most emissions per kg?”
Per 1000 kcal	Assess efficiency per energy delivered	“Which foods are most climate-efficient per calorie?”
Per 100g protein	Assess efficiency of protein production	“Which foods offer the lowest emissions per protein output?”

This structured approach ensures that the environmental impact of food production is thoroughly assessed, with clear pathways to evaluate ecological performance across food types and sustainability dimensions. It also facilitates consistent reporting and interpretation, helping stakeholders from policymakers to consumers make data-informed choices.

1.3 Hypothesis

To guide the analytical investigation, the following hypotheses were established to assess the comparative environmental impact of animal-based and plant-based foods:

- **Null Hypothesis (H_0):** There is no statistically significant difference in average environmental impact (GHG emissions, land use, water consumption) between animal-based and plant-based foods.
- **Alternative Hypothesis (H_1):** Optimized On average, animal-based foods have a significantly higher environmental impact across GHG emissions, land use, and water consumption compared to plant-based foods.

These hypotheses provide a structured foundation for statistical testing and support evidence-based conclusions regarding the sustainability of different food categories.

1.4 Key Business Questions

The analysis is structured around critical business questions that guide insights and decision-making:

1. **What is the overall environmental impact of animal-based vs. plant-based foods?**
Purpose: Determine whether plant-based foods are more sustainable than animal-based foods.
2. **Which food categories contribute the most to greenhouse gas emissions?**
Purpose: To Identify the worst offenders in terms of climate impact to promote eco-friendly diets.
3. **How do different supply chain stages contribute to emissions and land use?**
Purpose: Pinpoint which stages (e.g., farming, transport, processing) require intervention.
4. **What are the trade-offs between land use, water consumption, and emissions?**
Purpose: Assess if foods with low emissions have high water use, and vice versa.

5. **How do specific high-impact foods (e.g., beef, dairy, nuts) compared to more sustainable alternatives?**

Purpose: Provide alternatives for high-impact foods to guide sustainable consumption.

6. **Which foods are most affected by water scarcity, and how does this compare to their nutritional value?**

Purpose: Identify foods with high scarcity-weighted water use that may need water-efficient farming practices.

7. **How do different foods contribute to eutrophication, and what are the main drivers?**

Purpose: Understand how nutrient runoff from agriculture (e.g., nitrogen and phosphorus use) impacts freshwater ecosystems.

8. **Which food products exert the highest pressure on freshwater resources?**

Purpose: Identify foods with excessive freshwater withdrawals and assess their sustainability.

By structuring the analysis around these questions, the project builds a logical progression toward validating or rejecting the null hypothesis.

2.0 Methodology

A structured analytical approach was adopted to ensure alignment with the project's goal of understanding the environmental impact of food production. The methodology involved in-depth exploratory data analysis, robust feature engineering, and the preparation of data for interactive dashboard visualization. These steps enable transparent communication of insights to a broad range of stakeholders, including consumers, policymakers, and sustainability analysts.

2.1 Exploratory Data Analysis (EDA) and Data Cleaning

This stage focused on understanding the dataset's structure, addressing data quality issues, and transforming variables for analytical readiness.

2.1.1 Data Quality Assessment, Feature Engineering and Exploration

- **Assess Data Structure:** Used `.info()`, `.head()`, and `.describe()` to evaluate dataset dimensions, data types, and summary statistics.
- **Duplicate Removal:** Identified and removed redundant records to preserve data integrity.
- **Data Consistency Validation:** Verified that feature values fell within expected ranges (e.g., feature units, categorical groups).
- **Missing Value Review:** Confirmed data completeness and validated assumptions of no missing values or addressed them through structured methods.
- **Standardization of Column Names:** Long and verbose names were standardized using concise, consistent naming conventions (e.g., "Greenhouse gas emissions per 1000kcal (kgCO₂eq per 1000kcal)" → GHG_1000kcal).
- **Product Categorization:** Food products were classified into **Animal-Based Foods** and **Plant-Based Foods**, with further subcategories (e.g., Red Meat, Dairy, Vegetables).
- **Category Assignment:** Used mapping logic to assign each food item to corresponding Food_Category and Food_Subcategory, forming the structural foundation for subsequent analysis.

2.1.2 Univariate Analysis

- **Numerical Features:** Explored the distribution of environmental metrics (e.g., Total_emissions, Water_kg, Land_use_Change) using histograms and box plots to assess skewness and variability.
- **Categorical Features:** Analyzed frequency distributions of food categories and subcategories to understand the makeup of the dataset.

2.1.3 Bivariate Analysis

- **Category-Based Distributions:** Box plots and bar charts revealed patterns in environmental impact between animal- and plant-based products.
- **Top Contributors:** Created ranked bar charts to display the top 10 food products for each environmental metric based on percentage contribution.
- **Heatmaps:** Constructed subcategory-level heatmaps to visualize average impact per feature, supporting comparison across food groups.
- **Correlation with Food Type:** Quantified how strongly features correlated with Food_Category, identifying those most useful for differentiating between food types.

2.1.4 Multivariate Analysis

- **Correlation Heatmaps:** Showcased relationships among sustainability indicators, aiding in variable reduction and interpretation.
- **Pair Plots:** Visualized feature interactions with hue='Food_Category' to highlight how environmental variables cluster between animal-based and plant-based foods.

2.1.5 Handling Missing Values and Advanced Feature Engineering

- **Initial Structure Review:** Dropped unnecessary columns like Food_Category_num and verified data types.
- **Row-wise Missing Value Handling:**
 - Calculated the number and percentage of missing values across descriptive columns for each row.
 - Dropped 10 rows that had 30% or more missing data, ensuring the retained dataset remained robust and reliable for analysis.
- **Imputation Summary:**
 - Created df_aqinfo to track missing values, data types, and imputation strategies.
 - Performed **grouped median imputation** (e.g., Water_100gP by Food_Subcategory).

- Applied **regression-based imputation** for interdependent features using complete-case models.
- Adopted **distribution-aware methods** (mean or median) for skewed metrics.
- **Post-Imputation Validation:** Ensured all imputed features were now complete and checked consistency across all records.

As part of the data refinement process, a threshold of 30% missing values per row was applied to ensure analytical integrity. Based on this criterion, 10 out of 43 food products were removed from the dataset prior to conducting the analytical questions and hypothesis testing. These excluded items include *Wheat & Rye (Bread)*, *Maize (Meal)*, *Barley (Beer)*, *Tofu*, *Soymilk*, *Soybean Oil*, *Other Vegetables*, *Wine*, *Other Fruit*, and *Shrimps (farmed)*. Each of these food products had $\geq 30\%$ missing data across key environmental features, rendering them unsuitable for reliable comparison and statistical inference. While they were included during earlier stages of exploratory analysis (e.g., feature distribution and correlation heatmaps), they were excluded from all final comparative analyses, including the core hypothesis evaluation and business question answers. This step was essential to preserve the validity and robustness of the findings derived from the remaining 33 food products, which form the analytical backbone of the study.

2.2 Interactive Dashboard Development

Dashboards were designed using Power BI to provide an intuitive understanding of environmental trends and food system impacts.

- **Data Integration:** Processed and cleaned datasets were imported and structured into Power BI for dashboard creation.
- **Visualization Strategy:** Developed dynamic charts, filters, and maps to enable user-driven exploration of key metrics.
- **Stakeholder Engagement:** Visualizations were tailored for diverse audiences—policy advisors, sustainability analysts, and the public—supporting evidence-based decisions for promoting sustainable diets.



Figure 2. Illustration of the data refinement process, showcasing key steps from data quality assessment and duplicate removal to feature analysis and interactive dashboard development for delivering actionable insights.

3.0 Results

3.1 Univariate Analysis

3.1.1 Categorical Features

A. Food Category Distribution

- The dataset is dominated by Plant-Based Foods (76.74%), while Animal-Based Foods (23.26%) comprise a much smaller share.
- This imbalance suggests a stronger emphasis on plant-based sources within the data, possibly due to increased interest in sustainable diets.

B. Food Subcategory Distribution

- Among subcategories, Cereal Grains & Staples (16.28%) and Fruits (13.95%) are most prevalent, underscoring their dietary importance.
- These are followed by Vegetable Oils (11.63%), Vegetables (9.30%), and Legumes and Pulses, indicating widespread representation of core plant-based food groups.
- Animal-based subcategories such as Red Meat (9.30%) and Poultry and Seafood (4.65% each) are also represented, though less prominently.
- Less frequent categories include Beverages & Alcohol (2.33%).

C. Implications

- The dataset reflects a plant-forward dietary profile, likely shaped by a sustainability-focused analytical lens.
- The wide distribution of subcategories ensures diversity in product types, enabling more nuanced analysis in subsequent sections.

3.1.2 Numerical Features

A. General Observations

- Numerical features exhibit asymmetric distributions, with positive skewness and high kurtosis, indicating the presence of outliers and long-tailed distributions.
- Feature counts range between 26 and 43, confirming the presence of missing values across some metrics.
- The disparity between mean and median values further reinforces the non-normal nature of the data.

B. Carbon Emissions and Land Use

- **Total Emissions (mean = 5.97, median = 1.60, skew = 3.62)** Emissions are highly skewed, with a small group of products contributing disproportionately to the total footprint.
- **Farm Stage (mean = 3.47)** is the most dominant emission source among all supply chain stages.
- **Land-Use Change Emissions (mean = 1.26, skew = 3.69)** exhibits strong skewness and high variability, reinforcing that deforestation or land conversion contributes to emissions unevenly across food products.
- **Packaging, Processing, Transport, and Retail** stages have minimal contribution in comparison, indicating the primary impact lies in upstream agricultural practices.
- **Land Use (mean = 29.26, skew = 3.86)** The wide range in land requirements suggests that some foods (e.g., red meat) require significantly more space per kilogram.
- **GHG_1000kcal (mean = 5.63, skew = 3.36)** and **GHG_100gP (mean = 13.52, skew = 3.12)** These metrics confirm that certain protein-rich foods; notably beef, lamb, and dairy are among the highest GHG emitters per calorie and per gram of protein.

C. Water Consumption Trends

- **Water_kg (mean = 36,607.42, skew = 2.18)** Water consumption is unevenly distributed across products, with a few outliers (e.g., nuts, chocolate) consuming large amounts.
- **Water_1000kcal (mean = 17,380.58, skew = 0.68)** is less skewed, indicating that per-calorie water use is more evenly distributed.
- **Water_100gP (mean = 59,196.44, skew = 3.23)** shows extreme variability, highlighting that water use per protein is highly inefficient for some products, particularly fruits and nuts.

D. Eutrophication Potential

- **Eutro_kg (mean = 46.14, skew = 2.76)** Nutrient pollution is concentrated in a few products—primarily livestock and aquaculture—due to fertilizer use and manure runoff.

- **Eutro_1000kcal and Eutro_100gP** follow similar skewed patterns, though the per-protein distribution appears more balanced, suggesting that nutrient efficiency varies significantly by product.

3.1.3 Summary of Key Environmental Patterns

Category	General Trend	Key Takeaways
Land Use	Right-skewed, high variability	Disproportionate land demand from a few high-impact products
Water Use	Broad spread, extreme outliers	Certain foods (e.g., nuts, fruits) are extremely water-intensive
Eutrophication	Strongly skewed	Livestock and fertilizer-intensive crops drive nutrient runoff
GHG Emissions	Heavy-tailed, high outlier impact	High emissions driven primarily by animal-based products

Key takeaways

- The univariate analysis highlights clear disparities in environmental impact across food types, especially between plant-based and animal-based products.
- The distributions of emissions, land use, and water usage indicate that a few high-impact foods significantly influence overall sustainability metrics.
- These insights set the stage for deeper comparative analysis to validate hypotheses and explore targeted sustainability interventions.

3.2 Bivariate Analysis

3.2.1 Environmental Impact Comparison by Food Category

A. Higher Environmental Burden from Animal-Based Foods:

- Animal Feed, Farm Emissions, and Land Use Change Emissions are significantly higher in Animal-Based Foods, highlighting the disproportionate environmental cost of livestock farming.
- Total Emissions and GHG Emissions (both per kg and per kcal) are markedly higher for animal products.

B. Resource Intensity and Efficiency:

- Water Usage (per kg and per 1000 kcal) is considerably greater for animal-based foods, particularly dairy and meat products.
- Land Use (per kg, per kcal, per 100g protein) reveals that red meats require significantly more land for the same nutritional output.
- Eutrophication Indicators show greater pollution potential from animal-based agriculture due to manure runoff and fertilizer use.

C. Supply Chain Contributions:

- Emissions from Processing, Transport, and Packaging tend to be higher in animal-based foods, reflecting energy-intensive supply chains.
- Retail emissions are low across the board but slightly elevated for refrigerated goods such as meat and dairy.

3.2.2 Top Contributors by Metric

A. Land Use Change (Emissions):

- Beef (beef herd) leads land-use change emissions (30.07%), followed by Dark Chocolate (26.38%) and Cheese (8.30%).
- This reflects the GHG emissions from deforestation or land transformation, especially in livestock grazing and tropical crop plantations like cocoa and coffee.

B. Animal Feed Emissions:

- Pig Meat (14.87%), Beef (dairy herd) and Shrimps (farmed) (both 12.82%) dominate feed-related emissions.

- Feed production is a major upstream driver of emissions in animal agriculture.

C. Farm Stage Emissions:

- Beef (beef herd) (26.41%), Lamb & Mutton (13.07%), and Beef (dairy herd) (10.52%) contribute the most.
- These values underscore the direct on-farm GHG emissions from livestock.

D. Processing, Transport and Packaging:

- Palm Oil and Beef (beef herd) lead processing emissions (11.93%).
- Cane Sugar (9.52%), Beet Sugar (7.14%), and Olive Oil (5.95%) dominate transport emissions, showing food miles matter for some plant-based goods.
- Coffee (13.79%) tops packaging emissions, likely due to multilayer storage needs.

E. Retail Emissions:

- Highest for Beer, Soymilk, Tofu, Milk, and Cheese (10%), related to refrigeration.
- Fresh meat products like Beef and Poultry show moderate but consistent retail emissions (6.67%).

F. Total Emissions (kg CO₂-eq):

- Beef (beef herd) (23.21%) remains the single largest contributor.
- High contributions from Lamb & Mutton, Cheese, Dark Chocolate, and Coffee reinforce that some plant-based items also have intensive footprints.

G. GHG per 1000 kcal:

- Coffee (27.40%), Beef (beef herd) (19.60%), and Lamb & Mutton (6.74%) top the list.
- Calorically dense yet low-efficiency foods like Tomatoes (5.92%) also appear.

H. GHG per 100g Protein:

- Dark Chocolate (25.55%), Beef (beef herd) (13.66%), and Coffee (9.77%) dominate.
- Some low-protein, high-impact foods (e.g., fruits, chocolate) show elevated GHG per protein unit.

3.2.3 Land Use Insights (m²)

A. Total Land Use per kg:

- Lamb & Mutton (33.25%) and Beef (beef herd) (29.33%) top the list.

- Plant-based products like Chocolate, Coffee, and Oil Crops also require substantial land.

B. Land Use per 1000 kcal:

- Inefficiencies highlighted for Beef (29.15%), Lamb (28.46%), and Coffee (9.42%).
- Energy yield relative to land requirement is poor for some plant-based foods too.

C. Land Use per 100g Protein:

- Lamb (23.52%), Beef (20.82%), and Dark Chocolate (17.55%) require the most land per protein.
- More sustainable options include legumes, poultry, and fish.

3.2.4 Water Use Insights (L)

A. Water Use per kg:

- Highest: Cheese (15.82%), Nuts (11.66%), Farmed Fish (10.42%), and Rice (6.34%).
- Livestock and nuts require intensive water inputs.

B. Water Use per 1000 kcal:

- Farmed Fish (13.63%) and Tomatoes (12.87%) show poor water-to-energy conversion.
- Fruits and dairy continue to feature prominently.

C. Water Use per 100g Protein:

- Apples (16.06%), Berries (11.22%), and Tomatoes (8.99%) have poor water efficiency for protein.
- Cheese and nuts are also high.

3.2.5 Eutrophication Insights (g PO₄-eq)

A. Eutro_kg:

- Dominated by Beef (dairy herd) (20.83%) and Beef (beef herd) (17.19%).
- Farmed Fish (13.41%), Coffee (6.30%), and Chocolate (4.97%) follow.

B. Eutro_1000kcal and Eutro_100gP:

- Coffee (22.00%) and Dark Chocolate (12.22%) show high eutrophication per energy/protein.

- Nutrient-intensive farming of tropical and livestock-based products is a concern.

3.2.6 Scarcity-Weighted Water Use (L)

A. Per kg of Food:

- Nuts, Cheese, and Olive Oil dominate water stress impact.
- Animal agriculture (e.g., Lamb & Mutton, Beef, Pig Meat) remains a major contributor.

B. Per 1000 kcal and 100g Protein:

- Apples, Berries, and Brassicas show poor water efficiency for nutritional return.
- Nuts and dairy remain consistently intensive across all metrics.

3.2.6 Subcategory-Level Environmental Comparison

Food Subcategory	High Impact Areas	Interpretation
Red Meat	Land Use Change, Farm, GHG Emissions, Total Emissions	High impact across multiple sustainability metrics.
Dairy	Animal Feed, Total Emissions, GHG_100gP	Significant but generally lower than red meat.
Fruits	SW_Water_1000kcal, SW_Water_100gP	Low in emissions but water-intensive per protein.
Nuts & Seeds	Water_kg, SW_Water_100gP	High water use; requires efficient irrigation.
Seafood	Animal Feed, Water Use, GHG Emissions	Environmental pressure from feed and aquaculture systems.
Cereal Grains	Balanced impact across features	Relatively sustainable.
Beverages and Alcohol	Packaging	Low emissions, some packaging-related concerns.

3.2.7 Correlation Analysis with Food Category

The table below summarizes the correlation strength between environmental metrics and Food Category (Animal-Based = 1, Plant-Based = 0):

Correlation Range	Feature	Interpretation
Strong Positive (≥ 0.55)	Animal_Feed (0.91), Farm (0.55), Retail (0.56), Total_emissions (0.55), Eutro_kg (0.64)	Strongly associated with animal-based products.
Moderate Positive (0.30–0.50)	Water_kg, Water_1000kcal, Land_kg, Land_1000kcal, Processing, SW_Water_kg, Eutro_1000kcal, GHG_1000kcal, Land_use_Change	Generally higher in animal-based foods, but with some overlap.
Weak or Negligible (≤ 0.15 or negative)	Transport, Packaging, GHG_100gP, Water_100gP, SW_Water_100gP	Not strongly tied to food origin; some water metrics skew toward plant-based foods with low protein density.

Key Takeaways

- Animal-based foods consistently show a higher environmental footprint across most sustainability metrics.
- The largest impacts arise from farm emissions, animal feed production, and eutrophication due to nutrient runoff.
- Land use, water use, and total GHG emissions are significantly greater for animal products.
- Plant-based diets offer a promising route to reducing environmental degradation.
- Some plant-based products (e.g., coffee, chocolate, and nuts) also have notable impacts when assessed per kilocalorie or per 100g protein.
- A nuanced perspective is necessary; sustainability should be assessed on multiple dimensions, not just food origin.

3.3 Multivariate Analysis

3.3.1 Strong Positive Correlations

- **Total Emissions** show very high correlation with:
 - **Farm** activities ($r = 0.97$)
 - **Land_kg** ($r = 0.83$)
 - **GHG_1000kcal** ($r = 0.70$)
 - **Animal_Feed** ($r = 0.57$)

Emissions are predominantly driven by on-farm activities and land use, with animal feed also playing a significant role. These processes should be the focal points for reducing the environmental impact of food systems.

- **Land_kg** and **Land_1000kcal** ($r = 0.97$) indicates consistent land pressure across both mass-based and energy-based perspectives.
- **Water_kg** and **SW_Water_kg** ($r = 0.85$) suggests that surface water use is a major component of total water use, with similar trends across food types.
- **Eutro_kg** has strong correlations with:
 - **Farm** ($r = 0.77$)
 - **Total_emissions** ($r = 0.76$)
 - **Processing** ($r = 0.61$)

Suggests eutrophication is influenced heavily by early production stages.

3.3.2 Weak or Negative Correlations

- **Transport** exhibits low or negative correlations with other features. This implies that transport emissions are relatively decoupled from the production processes.
- **Packaging** shows weak to negative associations with sustainability metrics
 - E.g., **SW_Water_1000kcal** ($r = -0.27$), **Water_1000kcal** ($r = -0.29$) indicates a relatively minor environmental contribution in the dataset.
- **Retail** is weakly correlated across all metrics
 - Highest observed correlation is with **Land_1000kcal** ($r = 0.52$). Downstream impacts such as retail have limited influence on total emissions and resource use.

3.3.3 Interesting Mid-Range Correlations

- **Processing** correlates moderately with:
 - **GHG_1000kcal** ($r = 0.49$)
 - **Eutro_kg** ($r = 0.61$) reflects a secondary but noteworthy contribution, particularly toward eutrophication.
- **SW_Water_1000kcal** is moderately correlated with:
 - **SW_Water_kg** ($r = 0.59$)
 - **Water_1000kcal** ($r = 0.66$) reinforces the reliability of using calorie-based water stress indicators.
- **GHG_1000kcal** and **Land_1000kcal** ($r = 0.65$). Emission-intensive foods tend to also demand more land per calorie, highlighting a shared environmental burden.

4.0 Discussion

4.1 Assessing the Climate Impact of Animal-Based vs. Plant-Based Foods

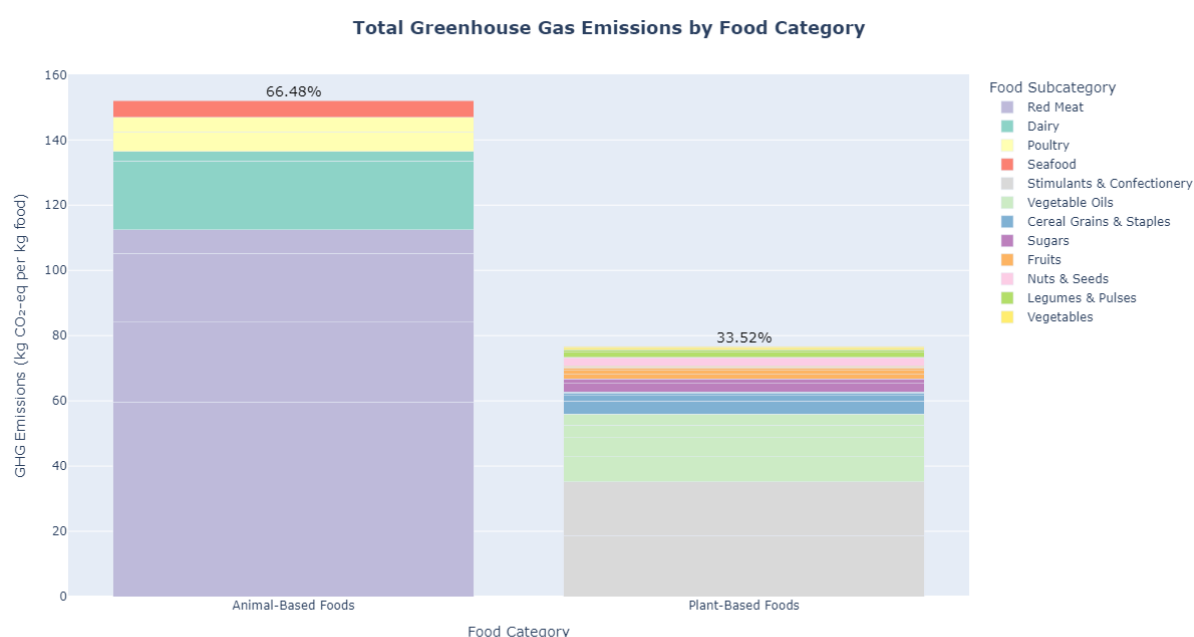


Figure 3. Comparison of greenhouse gas emissions (GHG) from animal-based and plant-based foods. The plot highlights the contribution of each food category to total emissions, emphasizing the higher environmental impact of animal-derived products, especially red meat, compared to plant-based options such as cereals, legumes, vegetables, and fruits.

The comparison between animal-based and plant-based foods reveals a stark contrast in their environmental footprints, especially concerning greenhouse gas (GHG) emissions. The analysis shows that animal-based foods account for 66.48% of total GHG emissions, more than double the 33.52% contribution from plant-based foods. This disparity underscores the disproportionately large climate burden carried by animal-derived products.

Within the animal-based category, red meat stands out as the most significant contributor, followed by dairy, poultry, eggs, and seafood. The high emissions from red meat are largely driven by methane release from ruminants, manure management, and feed production. These upstream processes intensify the climate impact associated with meat-heavy diets.

In contrast, plant-based food categories including cereal grains, legumes, vegetables, and fruits, consistently exhibit much lower GHG emissions per kilogram of product. While some plant-derived products such as nuts, chocolate, and coffee have higher impacts under specific metrics (e.g., per kcal or per gram of protein), their overall contributions to total emissions remain relatively low.

This disparity has critical implications for sustainability strategies. A transition toward plant-based diets, particularly those emphasizing legumes, grains, and vegetables can offer significant reductions in GHG emissions. Moreover, the data highlight that targeting reductions in red meat consumption alone could yield substantial climate benefits.

Observation	Implication
Animal-based foods contribute twice as much to GHG emissions as plant-based foods.	Promoting plant-forward diets can serve as a key climate mitigation strategy.
Red meat is the most intensive emitter across all food types.	Public awareness and policy should focus on reducing red meat consumption.
Legumes, vegetables, and grains have minimal climate impact.	These should be emphasized in sustainable dietary guidelines and food policies.

In conclusion, the analysis provides strong evidence supporting the environmental advantages of plant-based food systems. From a policy perspective, these findings can inform dietary recommendations, agricultural subsidies, and climate action plans aimed at reducing the carbon footprint of the global food supply.

4.2 Identifying the Highest and Lowest Greenhouse Gas Emitters

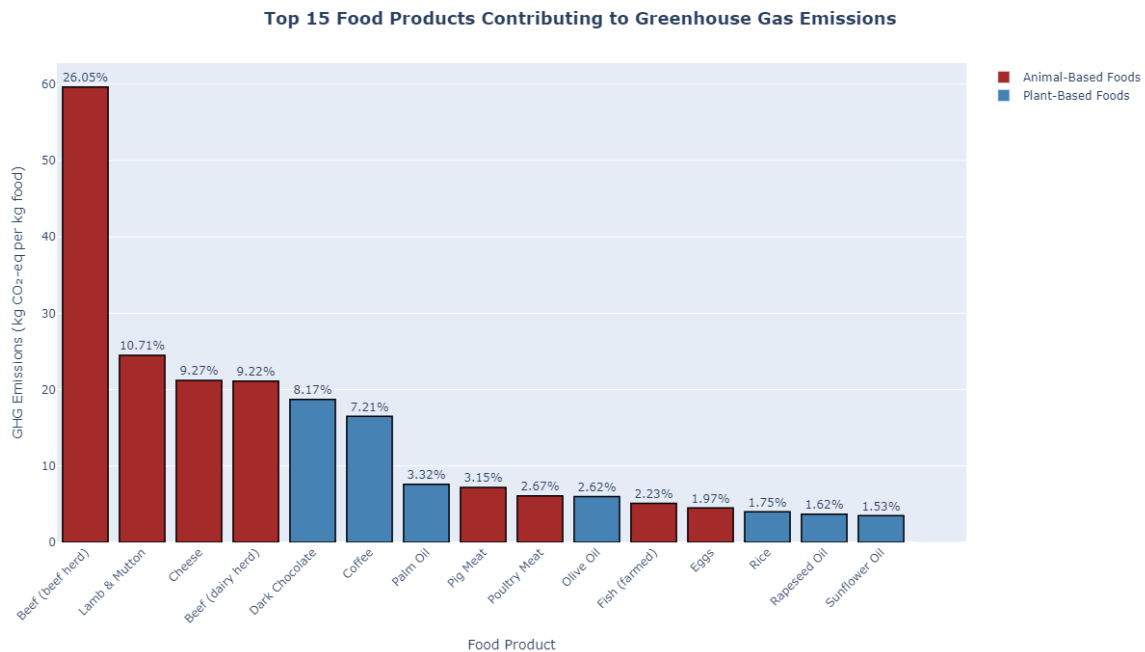


Figure 4. Top 15 Highest GHG Emitting Food Products (per kg food). This plot displays the food products with the highest greenhouse gas emissions per kilogram of food. Animal-based foods, particularly beef (from both beef and dairy herds), lamb & mutton, and cheese, dominate the rankings, with emissions ranging from ~23 to ~60 kg CO₂-equivalents per kg of food. Two plant-based products; dark chocolate and coffee, also appear among the top emitters ranging from ~16 to ~19 kg CO₂-equivalents per kg of food.

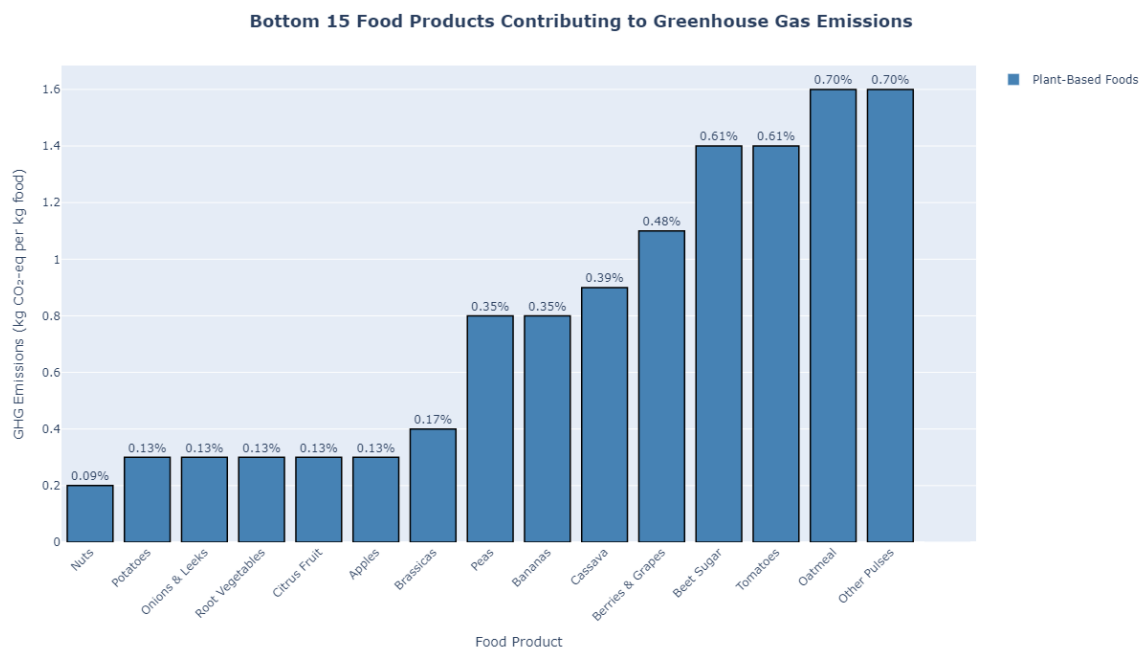


Figure 5. Top 5 Lowest GHG Emitting Food Products (per kg food). This plot highlights the lowest GHG-emitting foods per kilogram of food, all of which are plant-based. Foods like nuts, potatoes, onions, root vegetables, and citrus fruits produce less than 0.4 kg CO₂-equivalents per kg food, making them climate-efficient dietary options.

One of the central aims of this analysis was to determine which food products contribute most significantly to greenhouse gas (GHG) emissions, thereby identifying strategic targets for dietary and production-based sustainability interventions.

The findings reveal a clear dominance of animal-based foods among the highest GHG emitters. Notably, beef from the beef herd emerges as the single worst offender, contributing approximately 26% of total emissions overall. This is followed closely by lamb & mutton, cheese, and beef from dairy herds; all of which are animal-derived and intensive in terms of both feed production and methane emissions.

Interestingly, two plant-based products, dark chocolate and coffee, also feature among the top emitters, primarily due to land use change, fertilizer use, and energy-intensive processing in tropical production regions. Their inclusion reinforces the importance of assessing environmental impact on a product-specific rather than category-wide basis.

At the other end of the spectrum, the lowest GHG emitters are entirely plant-based, with foods like nuts, potatoes, onions, root vegetables, and citrus fruits emitting less than 0.4 kg CO₂-eq per kg of food. These products are nutrient-rich, calorie-efficient, and environmentally sustainable, making them optimal candidates for low-impact dietary recommendations.

Observation	Implication
Animal-based foods dominate the top GHG contributors.	Reducing intake of red meat and dairy could yield substantial reductions in carbon emissions.
Certain plant-based items (e.g., chocolate, coffee) rank high.	Life cycle emissions must be assessed holistically—not all plant-based foods are low-impact.
Low-emission foods are plant-based and nutrient-rich.	Potatoes, legumes, and vegetables offer scalable, eco-friendly alternatives.

In conclusion, this evidence strongly supports the promotion of climate-friendly diets, centred around plant-based foods. Public campaigns, supply chain restructuring, and dietary guidelines should prioritize reducing consumption of red meat and high-emission animal products, while also considering the hidden impacts of certain plant-derived commodities.

4.3 Supply Chain Emissions: Identifying Critical Stages for Intervention

Understanding where greenhouse gas (GHG) emissions originate within the food supply chain is essential to identifying intervention points for sustainability. The analysis reveals that emissions are highly concentrated in upstream processes, particularly farming and land use change, while downstream stages like transport and retail contribute minimally. The plot confirms that over 80% of total emissions are generated in the farm and land transformation stages alone, with animal-based production being the largest driver.

4.3.1 Breakdown of GHG Emissions by Supply Chain Stage

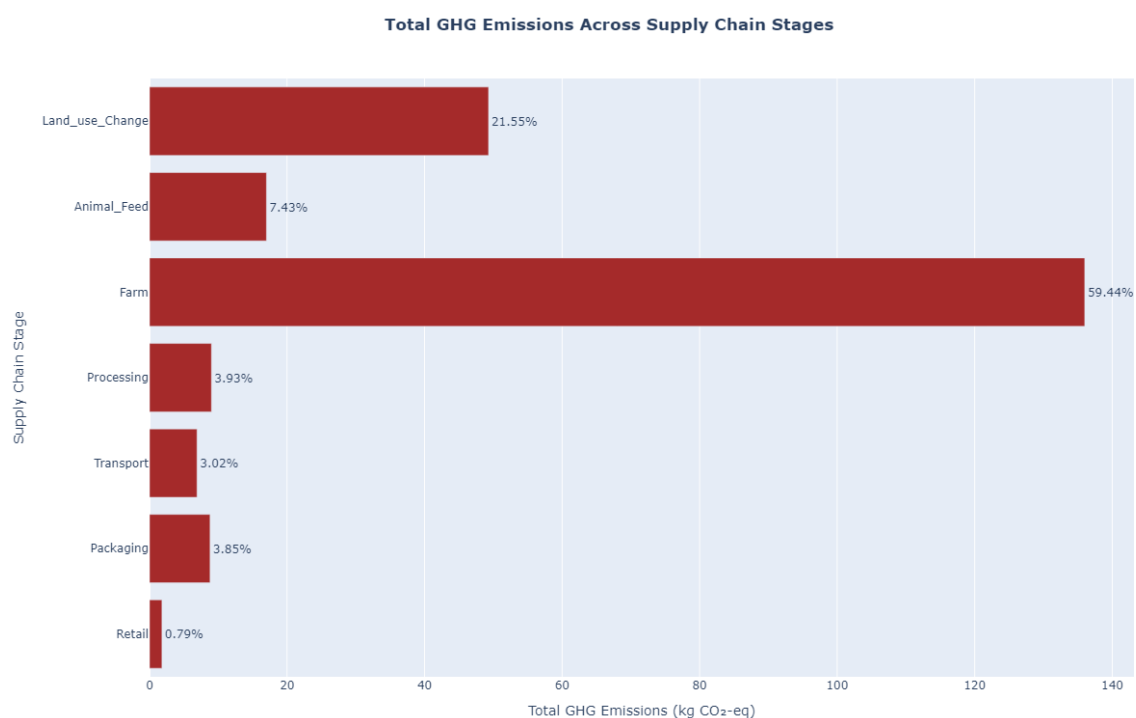


Figure 6. Proportional contribution of supply chain stages to total greenhouse gas (GHG) emissions across all food products. This chart illustrates the percentage share of each stage; farm, land use change, animal feed, processing, transport, packaging, and retail – in the total emissions footprint of the food system. The farm stage accounts for nearly 60% of total emissions, followed by land use change (21.55%) and animal feed (7.43%). Downstream processes contribute relatively minor shares, emphasizing the importance of upstream interventions to reduce environmental impact.

To reduce food system emissions effectively, targeted strategies are needed at the stages with the most impact:

- **Farm Stage (59.44%) – Highest Priority**
 - Encourage regenerative agricultural practices and precision farming.
 - Improve fertilizer application efficiency and manage manure more sustainably.
 - Promote low-emission livestock management techniques.

- Land Use Change (21.55%) – Critical
- Implement forest preservation policies and discourage agricultural expansion into natural habitats.
- Support reforestation, agroforestry, and the protection of carbon-rich ecosystems.
- **Animal Feed (7.43%) – Medium Impact**
 - Focus on feed optimization, including sourcing alternatives that require less land and water.
 - Improve the sustainability of feed crop cultivation.
- **Secondary Stages (Each <4%) – Efficiency Gains**
 - Processing and Packaging: Streamline energy use, adopt circular packaging solutions.
 - Transport and Retail: Optimize logistics, invest in low-emission cold chains, and adopt energy-efficient retail technologies.
- To understand emissions at the product level, two heatmaps were used to analyze GHG contributions by stage for both top 15 highest and bottom 15 lowest emitting foods.

4.3.2 Top Emitting Food Products: Intensive and Complex Supply Chains

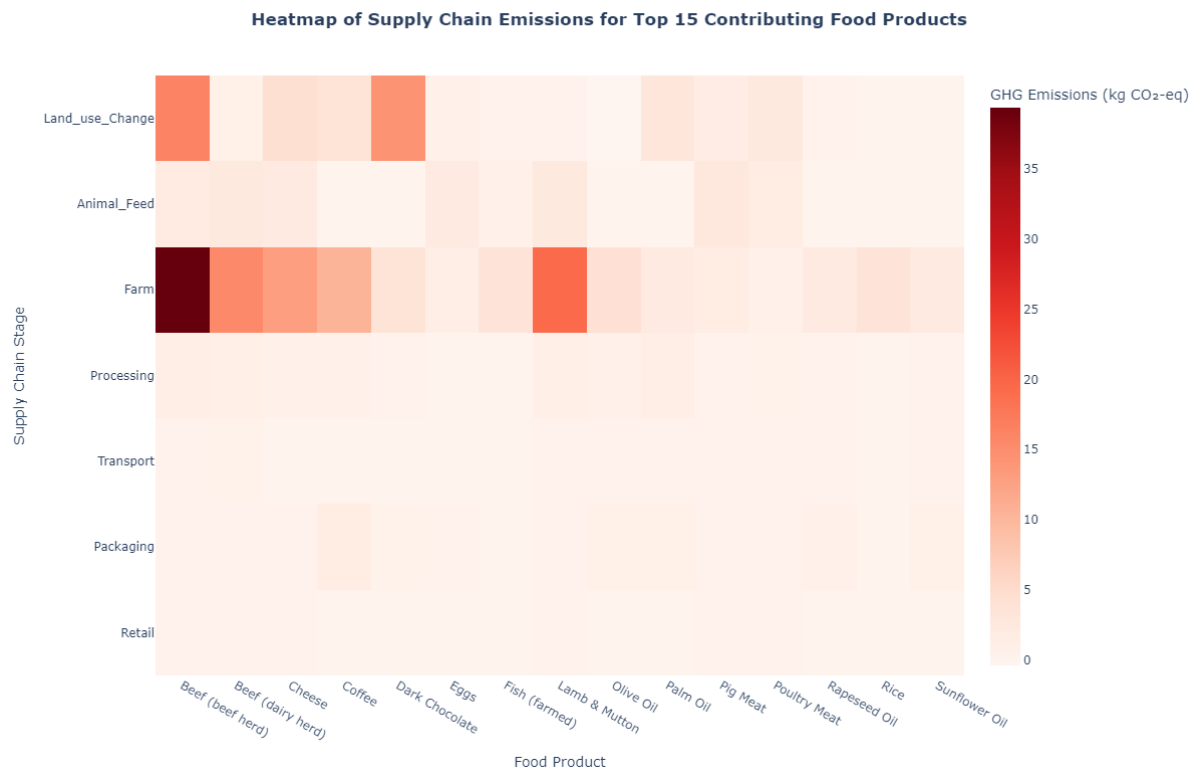


Figure 7. GHG emissions breakdown by supply chain stage for the 15 highest-emitting food products. This heatmap highlights emissions intensity across the supply chain for high-impact foods, primarily animal-based products and tropical commodities. The farm stage dominates, followed by significant contributions from land use change and animal feed. Downstream processes such as processing, packaging, and retail contribute marginally by comparison.

This heatmap shows GHG emissions across supply chain stages for the top 15 highest-emitting food products, which are mostly animal-based or tropical commodity crops. The key observations made include:

- The farm stage overwhelmingly dominates emissions, especially for ruminant meats (beef, lamb), driven by methane emissions, feed cultivation, and fertilizer use.
- Land use change is the second largest source, particularly for products associated with deforestation like beef, palm oil, chocolate, and coffee.
- Animal feed is a notable contributor for monogastric animals and aquaculture products such as pig meat, poultry, and fish.
- Processing and packaging contribute modestly for processed or shelf-stable items like coffee.
- Transport and retail remain marginal.

Supply Chain Stage	Overall Impact	Intervention Priority	Notes
Farm	Very High	Highest	Especially for beef, lamb, and cheese
Land Use Change	High	High	Especially for beef, palm oil, chocolate, coffee
Animal Feed	Moderate	Medium	Pig, poultry, fish, dairy
Processing	Low	Low	Slightly relevant for cheese, chocolate
Transport	Very Low	Low	Minimal contribution overall
Packaging	Very Low	Low	Slight uptick in processed goods
Retail	Very Low	Low	Negligible impact

4.3.3 Low-Emitting Food Product: Lean and Localized Systems

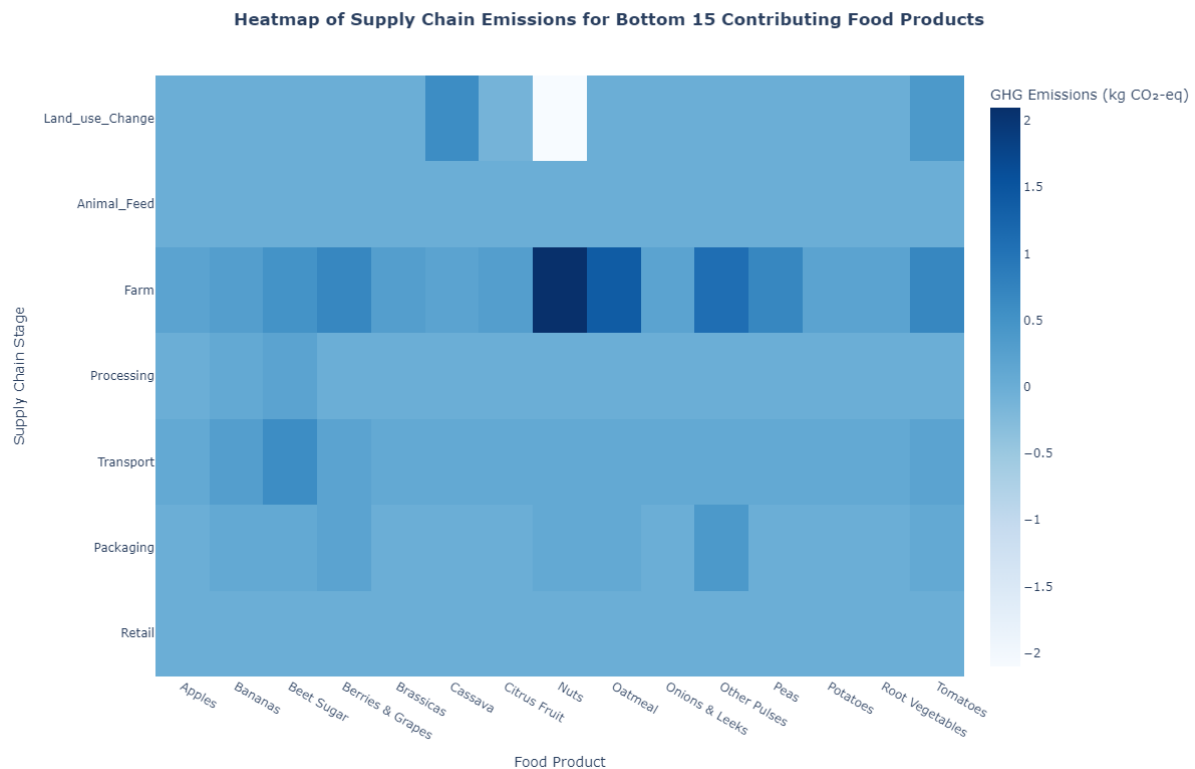


Figure 8. GHG emissions breakdown by supply chain stage for the 15 lowest-emitting food products. This heatmap visualizes the greenhouse gas contributions (kg CO₂-eq) from each stage of the food supply chain; farm for foods with the lowest total emissions. Although emissions are generally low, the farm stage remains the most significant contributor, while land use change, transport, packaging and downstream processes play a minimal role.

A separate heatmap analysis of the bottom 15 GHG-contributing food products (mostly plant-based) reveals important nuances:

- While these products are low emitters overall, the farm stage remains the dominant contributor.
- Transport emissions become slightly more relevant for perishable goods (e.g., bananas, berries, tomatoes) due to refrigeration and long shipping distances.
- Packaging and land use change are nearly negligible, and processing, and retail have almost no impact in this group.

Supply Chain Stage	Overall Impact	Intervention Priority	Notes
Farm	Moderate	Medium	Still the largest contributor, even in low-GHG foods.
Land Use Change	Very Low	Very Low	Negligible in most low-emission plant foods.
Transport	Low	Low–Medium	More relevant for imported or perishable items.
Processing	Very Low	Very Low	Minimal emissions.
Packaging	Very Low	Very Low	Slight spikes in some cases (e.g., pulses).
Retail	Very Low	Very Low	Uniformly insignificant.

The findings are clear: to significantly reduce GHG emissions in food systems, the greatest impact can be achieved by targeting upstream activities, especially farming practices and land use change prevention. While secondary stages like transport and packaging offer marginal opportunities for efficiency gains, their cumulative effect is small compared to upstream interventions.

However, even low-emitting foods are not exempt since they can still benefit from improved farm practices and optimized transport strategies, especially when dealing with fresh or globally sourced items.

This layered understanding of the food supply chain highlights the importance of tailored sustainability strategies emphasizing that not all stages, or all food types, require the same intensity of intervention.

4.4 Trade-offs Between Land Use, Water Consumption, and Emissions

This section evaluates the interrelationships among greenhouse gas emissions, land use, and water consumption across various food products. The aim is to determine whether foods with low emissions exhibit high water usage or land intensity, and vice versa – highlighting sustainability trade-offs that influence dietary choices.

4.4.1 Insights from Resource – Normalised Comparison (per Kilogram of Product)



Figure 9. GHG emissions (y-axis) versus land use (x-axis), with bubble size representing water use, plotted per kilogram of food product. This chart highlights sustainability trade-offs by comparing the climate, land, and water impacts of foods in their raw form. Plant-based foods tend to cluster in the lower-left quadrant (low emissions and land), while animal-based foods, especially red meat, exhibit higher environmental burdens across all three dimensions.

- **Animal-Based Foods Exhibit High Emissions and Land Use.** Animal-derived products, particularly red meats, impose a heavy environmental burden:
 - Beef (beef herd) exhibits the highest emissions (~60 kg CO₂-eq) and exceptionally high land use (~300+ m²/kg).
 - Lamb & mutton follow with significant land requirements (~370 m²/kg) and elevated emissions (~25 kg CO₂-eq).

- Cheese and beef from dairy herds also rank high in both metrics, albeit less severe than beef from beef herds.
- **Plant-Based Foods Cluster in the Sustainable Zone.** Most plant-derived foods (e.g., legumes, vegetables, grains) are in the bottom-left quadrant of the trade-off plot characterized by low emissions and low land use. However, water use varies within this group:
 - Nuts, while low in emissions and land use, often exhibit large water footprints, as indicated by the larger bubble size on the chart.
- **Evident Trade-Offs Across Sustainability Dimensions**
 - Nuts illustrate a classic trade-off: despite low emissions and minimal land use, their high-water requirements raise sustainability concerns.
 - Farmed fish demonstrate moderate emissions and land use with lower water consumption, making them a relatively better alternative among animal-based options.

4.4.2 Summary of Trade-offs (per Kilogram of Product)

- **Low emissions do not always equate to low water use.** Some plant-based foods (e.g., nuts) are water-intensive despite being climate-friendly.
- **Animal-based foods are generally high across all three dimensions:** land, emissions, and water use; underscoring their inefficiency.
- The most sustainable choices are foods that simultaneously maintain low emissions, low land use, and low water use, such as legumes, vegetables, and grains.

4.4.3 Insights from Energy-Normalized Comparison (Per 1000 kcal)

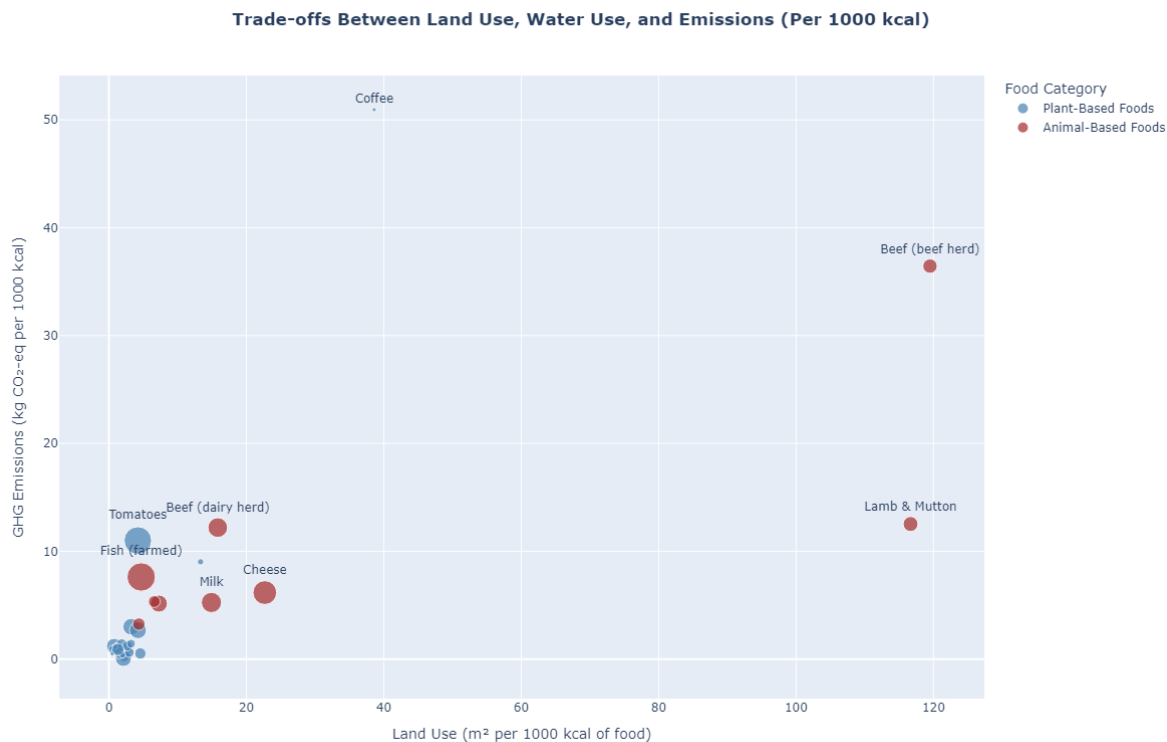


Figure 10. GHG emissions (y-axis) versus land use (x-axis), with bubble size representing water use, plotted per 1000 kilocalories of food. This energy-normalized chart reveals how low-calorie foods such as tomatoes and coffee can appear disproportionately unsustainable due to inflated land, emission, and water impacts per calorie. Conversely, calorie-dense plant foods like legumes and grains emerge as efficient sources across all metrics.

To account for differences in energy density, foods were also evaluated per 1000 kilocalories.

A. High Environmental Cost per 1000 kcal (Animal-Based)

Food	Land Use (m²)	Emissions (kg CO ₂ -eq)	Water Use	Insight
Beef (beef herd)	~120	~37	Medium	Extremely inefficient calorie source with large land and emissions burden.
Lamb & Mutton	~115	~14	Medium	Less carbon-intensive than beef, but still unsustainable.
Cheese	~25	~6.5	Large	High water use and moderate emissions.

Food	Land Use (m²)	Emissions (kg CO ₂ -eq)	Water Use	Insight
Milk	~15	~4.5	Large	Low calorie density inflates environmental cost per kcal.

Animal-based foods remain resource-intensive even when normalized by energy content.

B. Surprising Outliers (Low-Calorie Plant Foods)

Food	Land Use (m²)	Emissions (kg CO ₂ -eq)	Water Use	Insight
Coffee	~40	~51	Very low	High impact per kcal due to negligible energy contribution.
Tomatoes	~5	~1.2	Very large	Low energy density inflates water footprint per kcal.

Some foods appear unsustainable per calorie not due to poor practices but due to low caloric content.

C. Efficient Caloric Sources (Mostly Plant-Based)

Food Group	Land Use	Emissions	Water Use	Insight
Legumes, grains, roots	Low	Low	Low – Medium	Highly efficient calorie providers.
Farmed fish	Moderate	Moderate	Medium	More efficient than red meat, but less so than plant-based foods.

Starchy plant foods and legumes offer an optimal balance for sustainable energy provision.

4.4.4 Insights from Protein-Normalized Comparison (Per 100g Protein)



Figure 11. GHG emissions (y-axis) versus land use (x-axis), with bubble size representing water use, plotted per 100 grams of protein. This protein-normalized view emphasizes the inefficiency of red meats and certain dairy products, which require extensive resources for relatively small protein yields. Legumes, cereals, and pulses demonstrate high sustainability as protein sources.

A. High Environmental Cost per 100g Protein (Animal-Based)

Food	Land Use	Emissions	Water Use	Insight
Beef (beef herd)	Very high	~55 kg CO ₂ -eq	Medium	Extremely inefficient protein source.
Lamb & Mutton	Highest	~20 kg CO ₂ -eq	Medium	Worst land use efficiency per protein.
Cheese	Moderate	~15 kg CO ₂ -eq	Large	Dairy protein is carbon- and water-intensive.

Red meats and dairy are unsustainable sources of protein.

B. Notable Plant-Based Outliers

Food	Land Use	Emissions	Water Use	Insight
Dark chocolate	Very high	~90 kg CO ₂ -eq	Medium	High emissions and low protein content make it inefficient.
Coffee	High	~35 kg CO ₂ -eq	Very low	Poor protein source with high environmental cost.
Tomatoes	Medium	~20 kg CO ₂ -eq	Medium	Low protein density inflates environmental footprint.

Foods low in protein appears environmentally burdensome per 100g protein, but this is primarily a reflection of their nutritional profile rather than poor agricultural practices.

C. Efficient Protein Sources

Food	Land Use	Emissions	Water Use	Insight
Legumes, cereals, rice	Very low	Very low	Low – Medium	Protein-efficient and climate-friendly.
Peas	Very low	Very low	Low	Ideal source: minimal land, GHGs, and water demand.

Legume- and grain-based proteins deliver high sustainability per unit of protein.

D. Summary of Trade-offs per Metric

Category	Land Use	Emissions	Water Use	Efficiency (per 100g protein)
Red Meat	Very high	Very high	Moderate	Extremely inefficient
Dairy	Moderate	Moderate– High	High	Inefficient
Low-protein plants	Low– Medium	Moderate– High	Variable	Misleading (low protein content)
Legumes & grains	Very low	Very low	Low– Medium	Highly efficient

The findings reveal that no single metric sufficiently captures environmental sustainability. While shifting towards plant-based diets substantially reduces emissions and land use, caution must be exercised with water-intensive crops, especially in drought-prone regions. Foods like legumes, grains, and root vegetables provide the best balance across all three environmental dimensions. Conversely, beef and lamb exhibit the least favourable trade-offs, regardless of the normalization approach.

4.5 Comparative Analysis of High-Impact Foods and Sustainable Alternatives

This section examines how selected high-impact foods compare to more sustainable alternatives based on greenhouse gas (GHG) emissions, land use, and water consumption. The goal is to provide evidence-based guidance for dietary shifts toward more environmentally sustainable options.

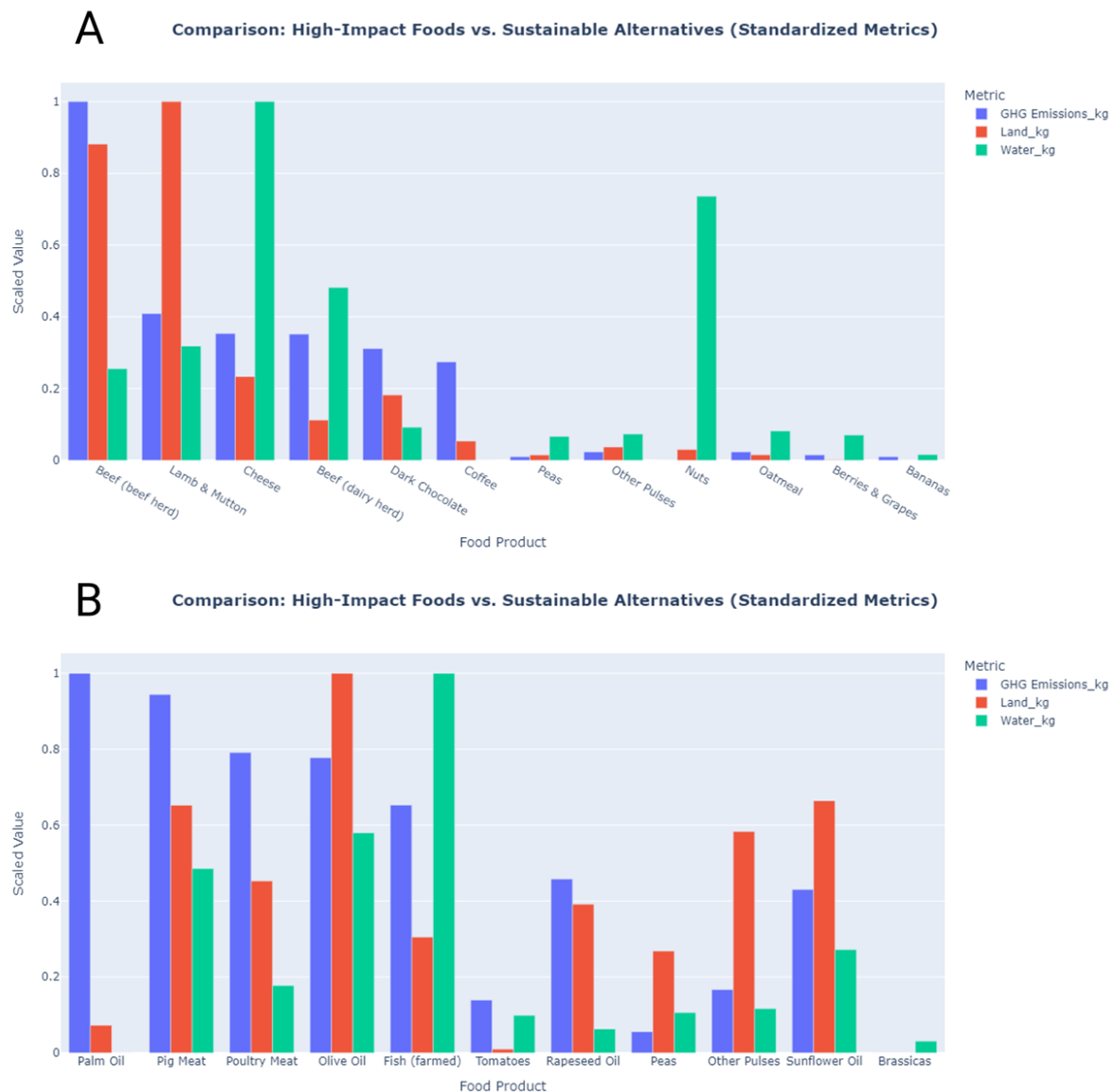


Figure 12. Comparison of high-impact foods versus more sustainable alternatives based on standardized environmental metrics. Each bar represents min-max scaled values for three key indicators: greenhouse gas (GHG) emissions per kilogram (blue), land use per kilogram (green), and freshwater use per kilogram (red). (A) The top panel highlights foods like beef, cheese, and dark chocolate with high environmental burdens, while the (B) bottom panel contrasts these with relatively sustainable options such as peas, pulses, and brassicas. This visualization emphasizes trade-offs across environmental domains to inform low-impact dietary choices.

4.5.1 Overview of Visualized Data

The visualizations use Min-Max scaled values to allow for meaningful comparisons across three environmental metrics:

- **GHG Emissions** (kg CO₂-eq)
- **Land Use** (m²/kg)
- **Water Use** (L/kg)

Food products are grouped into:

- **High-impact foods:** Beef (beef & dairy herd), Lamb & Mutton, Cheese, Dark Chocolate, Coffee, Palm Oil, Pig Meat, Poultry Meat, Olive Oil, Fish (farmed), and Tomatoes
- **Sustainable alternatives:** Peas, Pulses, Nuts, Oatmeal, Berries & Grapes, Bananas, Rapeseed Oil, Sunflower Oil, and Brassicas

4.5.2 Environmental Performance of High-Impact Foods

High-impact animal-based foods and certain oils exhibit the most adverse environmental profiles.

Food Product	GHG Emissions	Land Use	Water Use
Beef (beef herd)	Highest	Very High	Moderate
Lamb & Mutton	High	Highest	Moderate
Cheese	Medium	Moderate	Highest (1.0)
Beef (dairy herd)	Moderate	Moderate	Moderate
Palm Oil	Highest (1.0)	Very Low	Moderate
Pig Meat	Very High	Moderate	Moderate
Poultry Meat	High	Moderate	Low
Olive Oil	High	Highest	High
Fish (farmed)	Moderate	Low	Highest (1.0)
Dark Chocolate	Moderate	Low	Low

Food Product	GHG Emissions	Land Use	Water Use
Coffee	Low	Very Low	Very Low
Tomatoes	Low	Very Low	Very Low

Key insights include:

- Beef and lamb are the most unsustainable choices across all three metrics.
- Cheese exhibits an extreme water footprint despite moderate GHG and land impacts.
- Fish (farmed) presents high water demands, even with moderate GHG emissions.
- Palm oil stands out for its extremely high GHG impact despite low land usage.
- Tomatoes are a mild outlier in the high-impact group; overall low-impact but classified as high due to the comparative frame.

4.5.3 Performance of Sustainable Alternatives

In contrast, plant-based products—especially legumes, fruits, and vegetables—demonstrate superior environmental performance.

Food Product	GHG Emissions	Land Use	Water Use
Peas	Very Low	Very Low	Slightly Higher
Other Pulses	Very Low	Very Low	Slightly Higher
Nuts	Very Low	Very Low	High (~0.75)
Oatmeal	Very Low	Very Low	Low
Berries & Grapes	Very Low	Very Low	Low
Bananas	Very Low	Very Low	Very Low
Rapeseed Oil	Moderate	Moderate	Low
Sunflower Oil	Moderate	High	Moderate
Brassicas	Very Low	Very Low	Extremely Low

Key insights include:

- Brassicas, peas, and pulses offer highly sustainable profiles across all three dimensions.
- Nuts, though low in emissions and land use, require careful water management in water-scarce regions.
- Rapeseed oil emerges as a practical alternative to resource-intensive oils like palm or olive.

4.5.4 Business-Oriented Substitution Strategies

This comparison offers concrete substitutions for environmentally damaging foods:

High-Impact Food	Recommended Alternative	Justification
Beef (beef herd)	Peas / Pulses / Oatmeal	Over 10× lower GHG and land use
Lamb & Mutton	Bananas / Pulses	Significantly lower across all metrics
Cheese	Oatmeal / Pulses	Reduced water use and emissions
Palm / Olive Oil	Rapeseed Oil	Lower emissions, land, and water demands
Pig / Poultry Meat	Pulses / Brassicas	Substantially reduced overall impact
Fish (farmed)	Brassicas / Pulses	Avoids high water burden
Coffee	Bananas / Berries	Comparable enjoyment with lower footprint
Dark Chocolate	Berries	Similar sensory appeal with lower impact

4.5.5 The Case of Tomatoes: A Contextual Outlier

Tomatoes are an interesting anomaly in the high-impact group. While classified among high-impact foods due to their groupings, their actual environmental impact across GHG emissions, land use, and water consumption is relatively low compared to other members in the same category.

A. Tomatoes as a Sustainable Alternative

Given their low environmental footprint, tomatoes can serve as a replacement for certain high-impact foods:

- **Cheese (as a flavour enhancer):** Tomatoes, especially sun-dried or roasted, can provide umami depth in plant-based dishes without the water or GHG burden of cheese.
- **Processed meats (e.g., in sauces or spreads):** Tomato-based sauces can replace processed meat toppings in pizzas and pastas, reducing GHG emissions and saturated fat intake.
- **Palm oil / butter in stews or sauces:** Tomato purée or paste can substitute some of the richness provided by oils or butter while reducing environmental impacts.

B. Sustainable Alternatives to Tomatoes (if replacement is needed)

Although tomatoes have a low environmental footprint, in water-stressed regions or off-season conditions, they might not be ideal. Viable alternatives include:

- **Brassicas (e.g., broccoli, kale):** Excellent substitutes in cooked dishes and salads, offering nutritional value and a lower environmental footprint.
- **Berries & Grapes:** For flavour and antioxidant content in raw or salad applications.
- **Bananas or Peppers (in warm dishes):** These can provide body, colour, and sweetness in stews or sautés.
- **Peas (in sauces or blended form):** When pureed, peas offer a creamy texture and low-impact profile suitable for thickening sauces or soups.

4.5.6 Strategic Implications

- **For consumers:** Prioritize legumes, grains, fruits, and vegetables as primary protein and nutrient sources.
- **For businesses and retailers:** Reformulate product offerings and meal plan to reduce animal products and promote alternatives.
- **For policy makers:** Incentivize production and marketing of environmentally favourable foods, particularly legumes and Brassicas.
- **For regions with water constraints:** Promote low-water crops and educate consumers about the water implications of nuts and fish.

Replacing high-impact animal products and oils with plant-based alternatives can dramatically reduce environmental burdens. Legumes, Brassicas, and alternative oils like rapeseed emerge as key players in a sustainable food future.

4.6 Water Scarcity and Nutritional Value: Identifying Inefficient and Balanced Food Sources

Water scarcity is a growing concern in sustainable agriculture, especially as climate change exacerbates drought frequency and competition for freshwater intensifies. This section evaluates how scarcity-weighted water use per 100g of protein aligns with GHG emissions per 1000 kcal to identify foods that may require water-efficient farming innovations or dietary reconsideration.

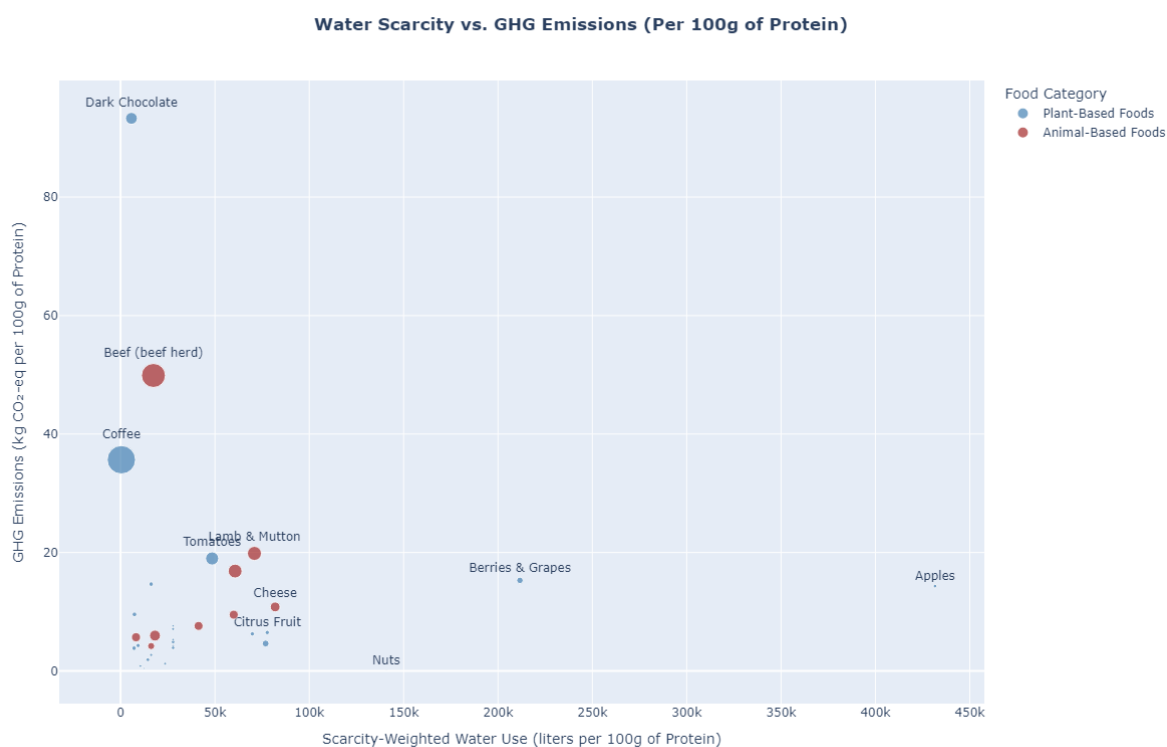


Figure 13. Scarcity-weighted water use versus greenhouse gas emissions per 100g of protein for selected food products. Each point represents a food item, with its position indicating its environmental impact in terms of water scarcity (x-axis) and GHG emissions per 100g of Protein (y-axis). Bubble size reflects GHG emissions per 1000 kcal, while colours distinguish between plant-based and animal-based foods. Notably, apples, berries and grapes, and nuts exhibit extremely high-water use per unit protein, while beef and dark chocolate show disproportionately high GHG emissions. The chart highlights trade-offs between carbon and water footprints across diverse food categories.

We visualize these trade-offs through a bubble chart where:

- The x-axis represents scarcity-weighted water use per 100g of protein.
- The y-axis represents GHG emissions per 100 grams of Protein.
- The bubble size indicates GHG emissions per 1000 kcal; highlighting the climate burden relative to energy delivered.

This framing allows us to assess both nutritional efficiency (protein per water used) and climate impact (GHG per calorie), drawing attention to foods with extreme resource burdens.

A. Key Observations

Food	Scarcity-Weighted Water Use	Bubble Size (GHG/1000 kcal)	GHG per 100g Protein	Remarks
Apples	Extremely high (~430k L)	Very small	~14	Low GHG per kcal and protein, but massive water use → nutritionally inefficient
Berries & Grapes	Very high (~220k L)	Small	~15	Low protein and moderate GHG makes them environmentally costly fruits
Nuts	Very high (~150k L)	Very small	~1	Excellent GHG profile, but high-water burden per protein unit
Dark Chocolate	High (~60k L)	Small – medium	~93	High GHG and water use, making it one of the least sustainable foods
Beef (herd)	Moderate (~20k L)	Large	~50	High protein content, but very high GHG emissions
Cheese	High (~80k L)	Small – medium	~12	Moderate GHG emissions with significant water use

Food	Scarcity-Weighted Water Use	Bubble Size (GHG/1000 kcal)	GHG per 100g Protein	Remarks
Coffee	Low water use	Large	~35	Not a protein source; high GHG per kcal makes it carbon intensive
Citrus Fruit	Moderate (~60k L)	Small	~6	Low GHG but not efficient in water use per protein
Tomatoes	Moderate (~50k L)	Medium	~18	Better water and GHG profile than most fruits
Lamb & Mutton	Moderate (~50k L)	Medium	~20	GHG and water impact are comparable to tomatoes, but from an animal source

B. Most Water-Inefficient Foods (per protein)

Apples, berries, and nuts stand out for their extreme water consumption per unit of protein. Despite low GHG emissions per kcal, their poor nutritional return on water investment marks them as inefficient from a sustainability standpoint. These foods may benefit from water-saving technologies or location-specific cultivation in low-stress regions.

C. Worst Overall (Water + GHG Impact)

Dark chocolate emerges as one of the least sustainable foods with very high GHG emissions paired with a considerable water footprint (~ 6k L), even after adjusting for its caloric and protein density. Beef (from the beef herd), while not the worst in water use, remains one of the highest GHG emitters, making it environmentally costly across multiple metrics.

D. Climate-Friendly but Water-Heavy

Nuts and apples illustrate a common trade-off in sustainability: low carbon emissions but disproportionate water footprints. Their production should be concentrated in regions with abundant water resources or enhanced with precision irrigation.

E. Balanced Trade-offs

Tomatoes represent a relatively sustainable option among fruits, with moderate water and GHG impact. Likewise, lamb and mutton, while more intensive than plants, show better profiles than beef, suggesting potential for more climate-conscious meat choices.

4.7 Eutrophication Potential and Climate Burden of Food Products

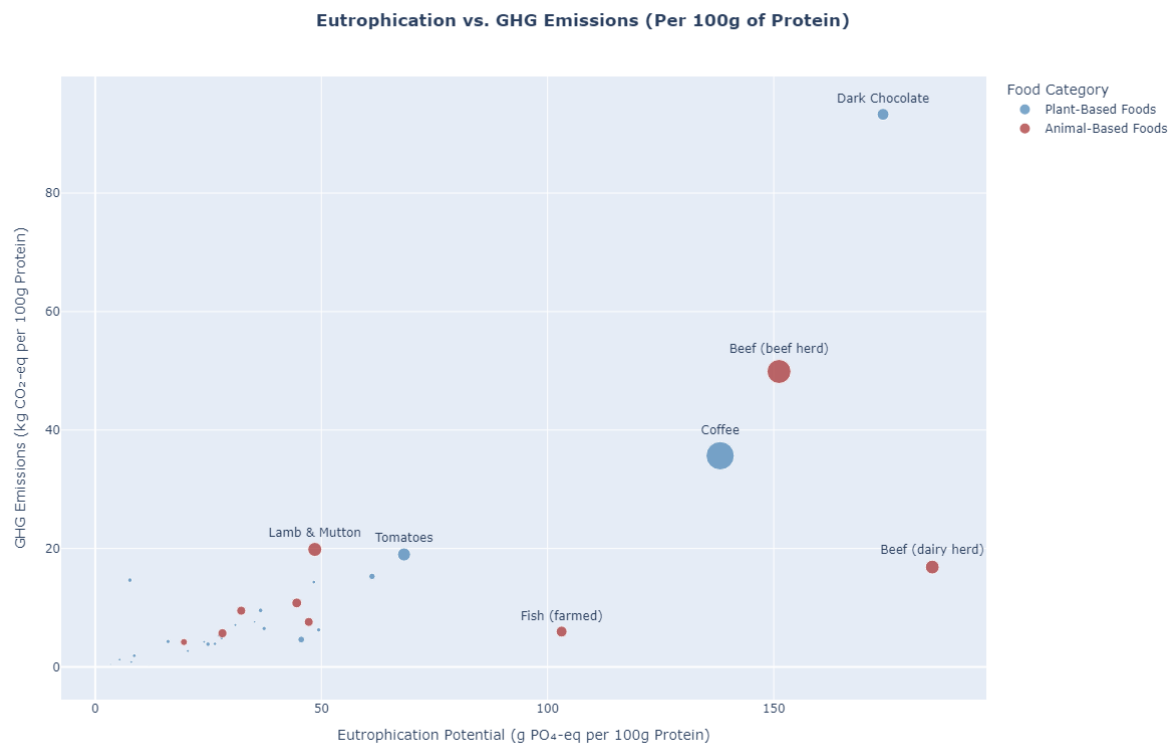


Figure 14. Eutrophication potential (g PO₄-eq) and greenhouse gas emissions (kg CO₂-eq) per 100g of protein for selected food items. Bubble size reflects GHG emissions per 1000 kcal, highlighting the climate burden relative to energy provided. Foods with high eutrophication potential and large climate burdens are clustered in the upper-right region, indicating poor overall sustainability.

Food	Eutrophication Potential (g PO₄-eq)	GHG Emissions (kg CO₂-eq)	Bubble Size (GHG/1000 kcal)	Remarks
Dark Chocolate	~160	~90	Small – medium	Extremely high eutrophication and GHG impact per protein; poor environmental profile overall
Beef (beef herd)	~145	~50	Large	Major contributor to both eutrophication and GHGs; large bubble reflects poor performance per kcal
Coffee	~135	~35	Largest	Not a protein source, yet very high eutrophication and climate burden per energy
Beef (dairy herd)	~180	~20	Medium	Very high eutrophication, lower GHG per protein than beef herd, still resource-intensive
Fish (farmed)	~100	~8	Small – medium	Lower GHG impact, but high eutrophication, possibly from feed and waste discharge

Food	Eutrophication Potential (g PO ₄ -eq)	GHG Emissions (kg CO ₂ -eq)	Bubble Size (GHG/1000 kcal)	Remarks
Tomatoes	~60	~18	Medium	Moderate eutrophication and relatively low GHGs make it more sustainable
Lamb & Mutton	~50	~20	Medium	Similar eutrophication to tomatoes, but slightly higher GHG emissions

- **High Eutrophication Drivers:**

- Foods like beef, dark chocolate, and coffee are clear outliers in terms of eutrophication potential.
- These values reflect nutrient pollution from feed crop fertilization, manure management, and waste from intensive cultivation practices.

- **Disproportionate Climate Impact per Calorie:**

- Coffee and beef (beef herd) show large climate burdens per kcal, meaning they generate considerable emissions without offering much caloric value.
- Dark chocolate, though with a moderate bubble, is unsustainable due to high impacts on both axes.

- **Relatively Lower Impact Foods:**

- Tomatoes, lamb and mutton, and farmed fish show a more favourable balance between protein delivery and environmental impact, though each has trade-offs.
- While farmed fish has low GHG emissions, its eutrophication potential still warrants concern.

Foods situated in the upper-right quadrant of the figure present the greatest environmental concern, combining high nutrient pollution and high GHG emissions per protein or calorie. Addressing eutrophication requires tackling agricultural runoff, feed crop management, and livestock waste systems. Shifting consumption toward lower-impact foods; particularly those with reduced nutrient runoff and more efficient protein or energy delivery could play a significant role in reducing pressure on aquatic ecosystems and climate systems alike.

4.8 Water Use and Climate Impact per 100g Protein

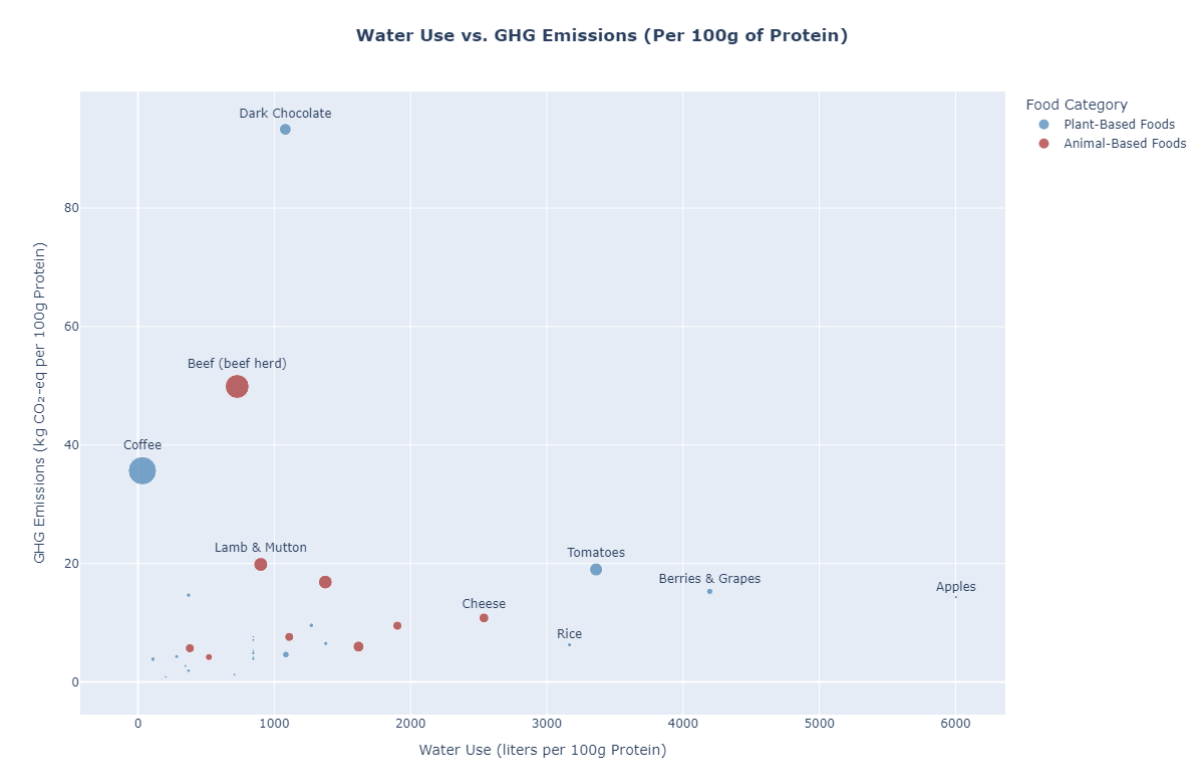


Figure 15. Comparison of water use (liters per 100g of protein) and GHG emissions (kg CO₂-eq per 100g protein) across various food products. Bubble size represents GHG emissions per 1000 kcal, emphasizing climate impact relative to caloric return. Foods with high water demand and large GHG burdens are highlighted as key contributors to environmental degradation.

Food	Water Use (L/100g Protein)	GHG Emissions (kg CO ₂ -eq)	GHG/1000 kcal (Bubble Size)	Remarks
Apples	~6000	~2	Very small	Extremely high water use per protein, low GHGs, low energy/protein content
Berries & Grapes	~4200	~10	Very small	High water use, moderate emissions, nutrient-light
Tomatoes	~3500	~15	Small–medium	High water demand, moderate GHGs, better nutrient density than fruits

Food	Water Use (L/100g Protein)	GHG Emissions (kg CO₂-eq)	GHG/1000 kcal (Bubble Size)	Remarks
Rice	~3000	~5	Very small	High water footprint, but low GHGs; a staple food with broad dietary relevance
Cheese	~2400	~12	Small	Significant water and GHG impacts; dense in calories and protein
Lamb & Mutton	~1800	~40	Medium	Balanced water use but very high GHGs per protein and kcal
Beef (beef herd)	~1100	~55	Large	Highest GHG emissions among animal products, moderate water use
Dark Chocolate	~1300	~90	Small – medium	Surprisingly high GHG emissions for a product with moderate water use
Coffee	~300	~35	Largest	Very low water use, but extreme climate impact per kcal due to low caloric return

- **Top Freshwater Stressors:**

- Fruits and vegetables like apples, berries, grapes, tomatoes, and rice consume enormous volumes of water per unit protein.
- Despite their low GHG emissions, their poor protein and energy density means that water is being expended inefficiently in terms of nutritional output.

- **Foods with Significant Environmental Trade-offs:**
 - Lamb & mutton show moderate water use but are heavy GHG emitters, with a medium-sized bubble indicating a notable climate burden per calorie.
 - Beef (beef herd) remains the most climate-intensive, with extremely high GHG emissions and a large climate bubble, despite using less water per 100g protein than most plant foods.
 - Dark chocolate and cheese continue to reflect dual concerns: moderate-to-high GHGs and notable water footprints, especially when judged against their nutritional benefit.
- **Outliers with Low Water Use but High Climate Impact:**
 - Coffee stands out again: it requires relatively little water but offers very little caloric value while producing excessive GHG emissions, making it a disproportionately harmful choice from a climate perspective.

Foods like apples, berries, tomatoes, and rice dominate in freshwater use per protein; a critical sustainability issue especially in water-scarce regions. Although these items are often plant-based and emit fewer GHGs, their low nutritional density raises questions about their long-term environmental efficiency.

On the other hand, animal-based products such as beef and lamb continue to rank high in climate burden, with cheese and dark chocolate also contributing significantly to total emissions.

Balancing water use and climate impact is essential. This means not just favouring plant-based diets indiscriminately, but prioritizing water-efficient, nutrient-dense plant foods while reducing intake of foods with excessive GHG emissions per kcal or protein.

4.9 Hypothesis Testing: Environmental Impact of Animal-Based vs. Plant-Based Foods

4.9.1 Mean Environmental Impact per Kilogram

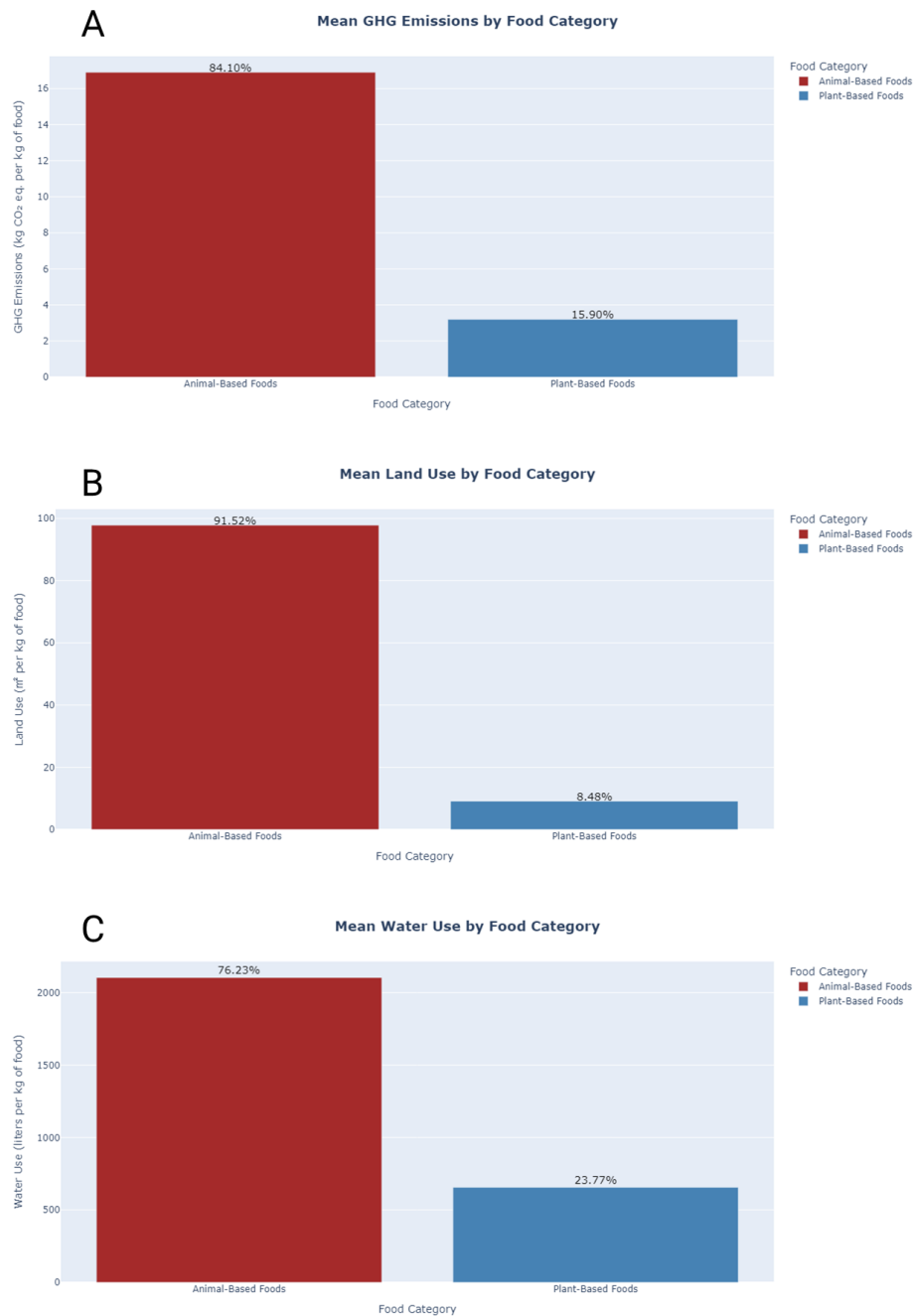


Figure 16. Comparison of average environmental impacts between animal-based and plant-based foods across three key indicators. (A) Mean greenhouse gas (GHG) emissions per kilogram of food, showing that animal-based foods contribute 84.1% of total emissions compared to 15.9% for plant-based foods. (B) Mean land use per kilogram of food, with animal-based foods using 91.5% of total land compared to 8.5% for plant-based foods. (C) Mean water use per kilogram of food, where animal-based foods account for 76.2% of water use, while plant-based foods contribute 23.8%. These results highlight the disproportionately higher environmental burden of animal-based food production.

These averages show that, per kilogram of product:

- Animal-based foods emit 5× more GHGs, require over 10× more land, and consume ~3× more water than plant-based foods.
- Animal-based products are therefore disproportionately responsible for environmental degradation per unit of output.

4.9.2 Statistical Testing Approach

To evaluate statistical significance, both parametric (t-test) and non-parametric (Mann-Whitney U) tests were considered. The final choice was guided by the following data characteristics:

- **Non-Normal Distributions:**
 - Skewness is notably higher in plant-based food distributions (e.g., GHG: 2.34 vs. 1.56).
 - High kurtosis in plant-based categories indicates heavy-tailed distributions (e.g., GHG: 4.48 vs. 1.5).
- **Heterogeneous Variance:**
 - Large disparities exist in variance across categories:
 - GHG: 327.47 (animal) vs. 23.39 (plant)
 - Land: 20,911.84 vs. 217.90
 - Water: 2,794,369 vs. 982,888
 - These differences violate the equal variance assumption of t-tests.

4.9.3 Test Results and Interpretation

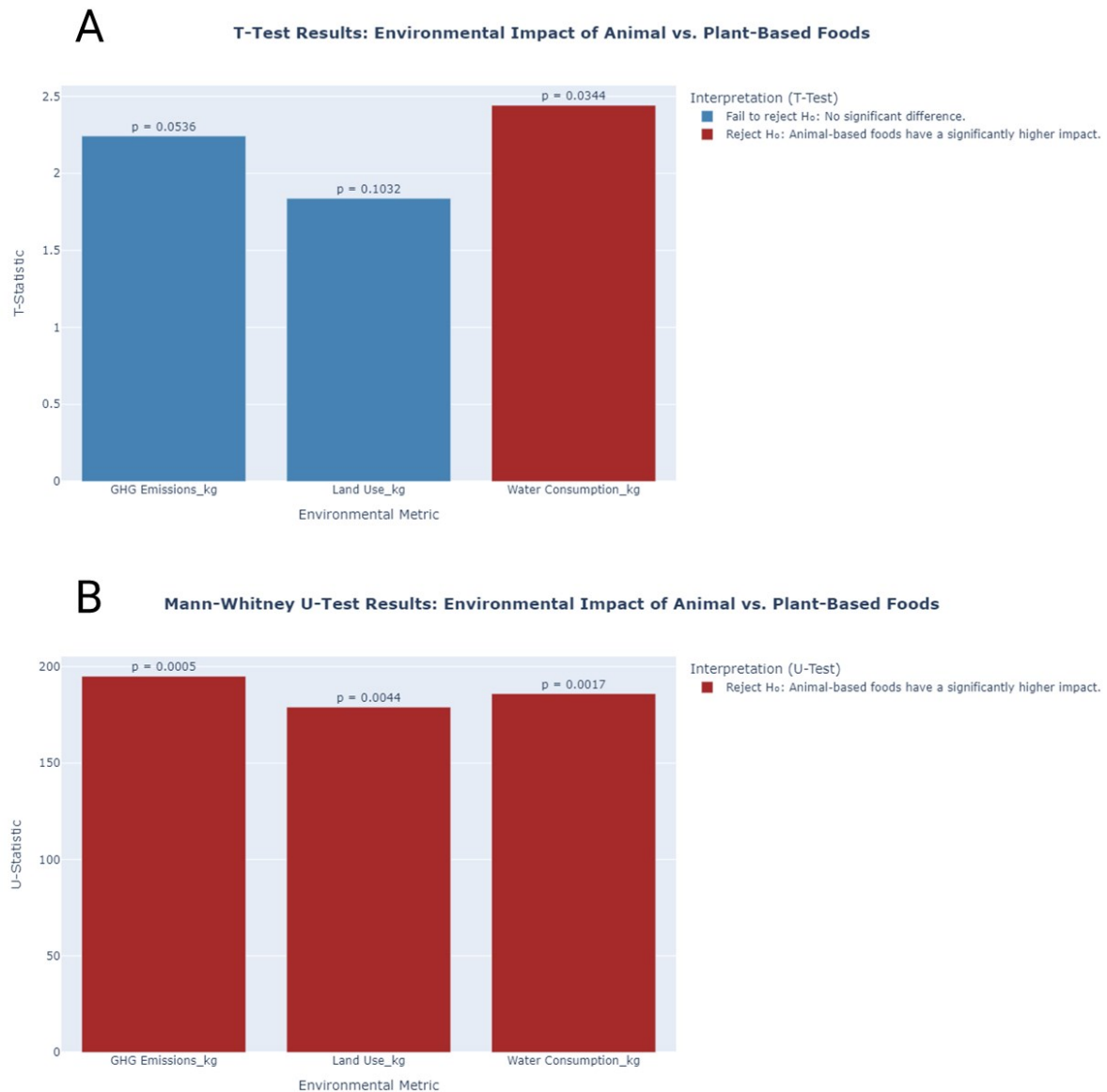


Figure 17. Statistical Comparison of Environmental Impacts Between Animal-Based and Plant-Based Foods. (A) Results from independent t-tests comparing mean greenhouse gas (GHG) emissions, land use, and water consumption. All per kilogram of food product. The bars are colored based on the interpretation: blue indicates no significant difference ($p > 0.05$), and red indicates a significant difference ($p < 0.05$). Only water consumption shows a statistically significant higher impact for animal-based foods. (B) Results from Mann-Whitney U-tests (a non-parametric alternative to the t-test) comparing the same environmental metrics. All three metrics per kilogram of food product; GHG emissions, land use, and water consumption, show statistically significant differences ($p < 0.05$), with animal-based foods having consistently higher environmental impacts. Together, these results suggest that animal-based foods generally have significantly greater environmental burdens compared to plant-based alternatives, especially in non-parametric comparisons.

These statistical tests show that, per kilogram of product:

- The t-test results are inconsistent, failing to reject H_0 in two of three metrics.
- In contrast, the Mann-Whitney U test shows consistently significant results across all three environmental metrics ($p < 0.05$).
- These findings confirm that the non-parametric Mann-Whitney U test is more robust for this dataset due to skewed, heavy-tailed, and heteroscedastic distributions.

Based on the Mann-Whitney U test, we reject the null hypothesis (H_0) for GHG emissions, land use, and water use. Therefore, we accept the alternative hypothesis (H_1): *"On average, animal-based foods have a significantly higher environmental impact across GHG emissions, land use, and water consumption compared to plant-based foods."*

This conclusion supports the broader narrative that animal agriculture imposes a substantially greater ecological burden, reinforcing the value of dietary shifts and agricultural policies aimed at sustainable food system transformation.

5.0 Conclusion

This study presents a comprehensive environmental impact assessment of 43 distinct food products, examining greenhouse gas (GHG) emissions, land use, water consumption, and eutrophication across both animal-based and plant-based categories. The results demonstrate that animal-based foods consistently exert a disproportionately high environmental burden, particularly across upstream supply chain stages such as farming, land use change, and animal feed production. In contrast, plant-based foods exhibit significantly lower emissions and resource demands, although certain products like coffee, chocolate, and nuts also reveal environmental concerns under calorie- or protein-normalized metrics.

Key findings highlight that:

- The farm stage contributes nearly 60% of total emissions, while land use change and animal feed collectively add another 30%, making these upstream stages critical intervention points.
- Beef, lamb, and cheese emerged as the most environmentally intensive products across all metrics.
- Red meat production demonstrates the highest emissions and land use demands per kg of food, per calorie, and per gram of protein.
- Plant-based foods, especially legumes, grains, and root vegetables, consistently appear as the most sustainable in terms of emissions, water use, and land efficiency.
- Water scarcity impacts were notably high for nuts, fruits, and cheese, suggesting the need to evaluate trade-offs even within plant-based options.

The use of heatmaps and categorical breakdowns has underscored the multi-dimensionality of environmental impact, emphasizing the importance of context-specific decisions rather than blanket assumptions.

6.0 Recommendations

Based on the insights gained from this analysis, the following evidence-based recommendations are proposed for various stakeholders:

For Policymakers and Regulators:

- Promote dietary shifts toward plant-based alternatives through subsidies, taxes, or public health campaigns, focusing especially on reducing red meat consumption.
- Enforce deforestation-free supply chains and incentivize land restoration projects, particularly in commodities like beef, palm oil, cocoa, and coffee.
- Develop policy frameworks for water-efficient agriculture, targeting high-scarcity foods like nuts, cheese, and fruits grown in arid regions.

For Agricultural and Food Industry Stakeholders:

- Invest in low-emission livestock practices (e.g., feed additives, regenerative grazing, methane digesters) to reduce farm-stage GHG emissions.
- Shift toward sustainable feed sourcing and precision feed management to lower animal feed-related emissions.
- Prioritize processing efficiency and sustainable packaging, especially for high-volume consumer goods.
- Optimize transport and cold-chain logistics for perishable items with elevated transport emissions (e.g., berries, bananas, tomatoes).

For Environmental Organizations and Sustainability Advocates:

- Focus advocacy on upstream interventions, such as sustainable farming and land conservation, where the greatest impact can be achieved.
- Educate consumers on nutrient-to-impact trade-offs, highlighting that not all plant-based foods are equally sustainable when normalized by protein or energy.

For Consumers:

- Make informed food choices that balance nutrition and sustainability, favouring legumes, grains, vegetables, and root crops.
- Reduce intake of high-impact animal products, especially beef and lamb, while being mindful of water-intensive alternatives.

- Support brands and certifications committed to sustainable sourcing and low-impact production systems.

These recommendations underscore the central message of this report: “Targeted interventions across food supply chains, particularly at the farm and land-use levels, hold the greatest potential to reduce environmental impact.” A collaborative approach involving consumers, industry, and policymakers is essential for transforming the global food system toward sustainability.

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